

The Motte-and-Bailey Framework for Leakage-Resilient Accordion Modes

Featuring Qaitbay and Alicante

Mario Marhuenda Beltrán^{1,2} 
and Mustafa Khairallah^{2,3} 

¹ Radboud University, Nijmegen, Netherlands
`mmarhuenda@cs.ru.nl`

² Nanyang Technological University, Singapore, Singapore

³ Umeå University, Umeå, Sweden
`khairallah@ieee.org`

Abstract. Accordion modes have experienced a surge in popularity, partially motivated by the recent NIST Accordion modes project. None of the existing practical constructions is leakage-resilient by default. In this work, we design a leakage-resilient Accordion mode. We start by presenting a generic analysis of the Encode-then-Encipher (EtE) framework in the leakage-resilient setting, assuming the enciphering is a leakage-resilient STPRP (STPRP ℓ 2). We show that the resulting security, while strong, suffers from some limitations. Next, we introduce **Motte-and-Bailey**, a general framework for building leakage-resilient accordion modes, in the spirit of the PIV construction. **Motte-and-Bailey**, or **MaB** for short, is a leveled construction, requiring light assumptions on most of its components to guarantee good STPRP ℓ 2, CIML2 and CCAM ℓ 2 security. In particular, we require two fully protected calls to a TBC, a collision-resistant hash function (with unbounded or light leakage), and an ideal leakage-resilient PRG, secure against single-trace attacks. Additionally, we present particular instantiations, **Qaitbay** and **Alicante**. In **Qaitbay** the PRG and the hash function are replaced by the Sponge function, while an independent TBC is used for the leak-free calls. **Alicante** makes use of an ideal cipher, and uses the MDPH hash function and the 2PRG construction, while the leak-free calls are implemented using independent calls to the ideal cipher. We also give three flavours of how to instantiate the TBC inside **Qaitbay**. Last but not least, we show how to strengthen **MaB**, **Qaitbay** and **Alicante** to also achieve CCAM ℓ 2.

Keywords: Accordion mode · AEAD · Leakage · Confidentiality · Integrity · EtE

1 Introduction

Over the years, symmetric-key encryption has seen multiple shifts in focus; from stream ciphers and Block Ciphers (BCs), through plain encryption modes, to

Authenticated Encryption with Associated Data (AEAD). These shifts have been motivated by attempts to build more robust notions, while keeping the encryption algorithms practical. Over the past two decades, intensive research on AEAD schemes [5] has been performed. For many years, AES-GCM [55] served as the primary AEAD standard. However, both performance [56] and security limitations [50] have been identified, motivating two major community-wide initiatives aimed at developing efficient and thoroughly analyzed AEAD schemes: the CAESAR competition⁴ and the NIST Lightweight Cryptography project⁵.

However, AEAD schemes have significant limitations. They rely on adding redundancy in the ciphertext to provide security. This leads to practical challenges. In disk/memory encryption, it may be hard to extend the plaintext size due to space limitations, which makes using AEAD infeasible. Similarly, applications that can extend the plaintext size may have hard time communicating these additional bits; the communication cost of sending the bits may be much higher than the computational cost. Because of this, proposals for short-tag or variable-length-tag AEAD schemes [31, 44, 61] have been considered, assuming that for many applications an occasional random forgery can be tolerated, as long as confidentiality is maintained. Unfortunately, it was shown recently that, for most efficient/cheap AEAD schemes, the relation between confidentiality and integrity is tight; after the first carefully constructed forgery, confidentiality is lost [45, 47].

These limitations are amongst the motivations to study a different type for symmetric key encryption: Tweakable Wide Block Ciphers (TWBCs). A TWBC is a keyed function that takes two inputs of arbitrary length. The first input is the tweak and the second input is the plaintext. The tweak is used to select a family of permutations over strings of length l bits, where $l_{\min} \leq l \leq l_{\max}$, such that for every plaintext of length l , the TWBC produces a ciphertext also of length l . The expectation is that each such permutation is indistinguishable from a random permutation. Such schemes ensure that any change in any bit of the plaintext or tweak affects the whole ciphertext, and vice versa. They provide security against chosen ciphertext adversaries, even if there are no integrity guarantees. Additionally, they can be easily adapted to AEAD using known techniques, *e.g.* the Encode-then-Encipher (EtE) framework [6].

The properties of TWBCs have motivated the National Institute for Standardization and Technology (NIST), USA, to start a new project with the goal of proposing a new standard. They renamed TWBCs as Accordion modes. This effort shows the pressing need for efficient and secure TWBCs [17].

1.1 Non-exhaustive exposition of techniques for building Accordion modes

The EtE framework was proposed by Bellare and Rogaway [6], assuming the existence of enciphering schemes. There have been many candidates for TWBCs over

⁴ <https://competitions.cr.yp.to/caesar-submissions.html> (2016)

⁵ <https://csrc.nist.gov/projects/lightweight-cryptography>

the years. The early works on building Pseudo-Random Permutations (PRPs) from Pseudo-Random Functions (PRFs) [53] can be seen as a basis for building TWBCs, and has indeed been used for such purpose [2]. Another early approach is Encrypt-Mix-Encrypt (EME) [40, 41], where the message is split into blocks, and a fixed-length BC is used to encrypt each block, then a mixing layer is applied on all the blocks before repeating the process again. In some sense, the whole scheme can be seen as a two-round Substitution-Permutation Network (SPN). A far more popular approach is the HCTR paradigm. This approach was first proposed by Wang *et al.* [67], and later popularized by the designs of **Adiantum** [20] and **HCTR2** [21]. Nevertheless, it was shown to have a security flaw where the hash function used was not almost-XOR-universal as required [49], and the security proof of [16] suffered from a bug [21]. **Adiantum** and **HCTR2** fix these issues. **THCTR** [28] was an attempt to enhance security, but the security analysis had a flaw that limited its gain [1, 46]. **HCTR+** [23] is a more recent candidate that avoids the flaws in **THCTR**. Cascaded **HCTR2** [18] is a very recent candidate that improves the security of **HCTR2** by applying the construction twice, with $1.5 - 2\times$ the cost. A different approach based on similar primitives has been proposed by Gunging *et al.* [32], and its efficiency was demonstrated by Dobraunig *et al.* [26].

Alternatively, the Protected IV (PIV) approach was proposed by Shrimpton and Terashima [64] as a modular approach for building TWBCs from smaller primitives. It has the advantage of being easy to understand and analyze, but early instances have proven more costly compared to HCTR variants. Recently, **GEM** [15] has been proposed. It is somewhat inspired by PIV, but not in a generic sense. None of the cited schemes achieve leakage resilience. To the best of our knowledge, no scheme in the literature achieves leakage resilience as a TWBC.

We note that the NIST has announced its plans to adopt variants of **HCTR2** as the new standard. **HCTR2** does not provide leakage resilience except if the implementation is heavily protected. This further motivates the necessity of our work, as there is no leakage-resilient Accordion mode with reasonable overhead.

1.2 Leakage-resilient symmetric cryptography

Side-channel analysis is an essential aspect of modern applied cryptography. Any computation executed on a physical device is inherently susceptible to unintentional information leakage through observable channels, such as timing variations, power consumption, or electromagnetic emanations. Power-based side-channel attacks are typically categorized into two main classes: Simple Power Analysis (SPA) and Differential Power Analysis (DPA). SPA attempts to extract information from a small number of traces, possibly 1, whereas DPA exploits statistical relations across a large number of traces collected under varying inputs. Here, the number of traces denotes the number of distinct inputs used in the acquisition process. However, repeated measurements of identical inputs are often employed to suppress noise. Analogous classifications apply to other forms of side channels. Countermeasures against such attacks frequently rely on physical protections, among which masking represents a central approach. The robustness of such schemes is characterized by their security order, with higher-order

masking offering increased resistance. Nevertheless, the associated implementation cost grows substantially with the masking order. A different, more recent approach, is to design the modes to be leakage-resilient. Early works in this area in the context of symmetric key cryptography started in 2008 [29], and have seen significant progress over the years both on the theoretical (c.f. [27, 30]) and practical (c.f. [24, 42, 52, 54, 60, 62, 65, 66]) sides. Most of the efforts have been dedicated to designing leakage-resilient AEAD, with the efforts converging towards what is now known as leveled implementations. This approach is explored in depth by Bellizia *et al.*, summarizing the state of the art at the time [8]. The main approach is that the mode is divided into components. A constant number of fixed-size components come with strong assumptions, the most popular of which is being leak-free. These assumptions typically mean that the implementation of these components is protected against side-channel with expensive countermeasures, such as higher-order masking. The rest of the components come with weaker assumptions, such as having unbounded leakage (they leak all their internal values), or hard-to-invert leakage as long as each secret input is never repeated, which translates to cheaper countermeasures against SPA. This approach has seen significant success in the context of AEAD, due to the observation that masking countermeasures are typically extremely expensive, adding several orders of magnitude compared to unprotected implementations. By ensuring that these countermeasures are used only in a handful of components, the overall implementation is significantly cheaper than if we mask the whole mode. Several AEAD designs have been proposed in this direction from (Tweakable) Block Ciphers (TBCs) [9, 24, 34, 36, 37, 48, 52, 62, 63]. Similar approaches have also been applied to permutation-based cryptography [7, 25].

When it comes to TWBCs, the situation is less clear. Two results have been published on the leakage resilience of PRPs. The first is by Dodis and Pietrzak [27], who showed that a Feistel cipher built from leakage-resilient round functions requires a super-logarithmic number of rounds to achieve leakage-resilient PRP security. However, this result is shown under two strong adversarial assumptions:

- The adversary is adaptive, not only in choosing the queries, but also the leakage functions.
- The PRFs used in the round functions are leakage-resilient, but the remainder of the Feistel network has unbounded leakage. In other words, all the “connections” and “additions” in the network leak their inputs and outputs.

The second result is due to Faust *et al.* [30], where they showed that against non-adaptive adversaries, three rounds are sufficient to reach PRP security (which is weaker than STPRP security). These two results are of theoretical nature, and pre-date a lot of the recent positive results on leakage-resilient AEAD.

1.3 Contributions

The paper can be divided into four main contributions:

1. Generic analysis of the leakage resilience of EtE when the underlying TWBC is leakage-resilient.
2. **Motte-and-Bailey (MaB)**: a dedicated framework for building leakage-resilient leveled TWBCs, and for which we can prove better EtE properties than the generic ones.
3. **Qaitbay**: an instantiation of MaB using Sponge functions.
4. **Alicante**: an instantiation of MaB using ideal ciphers.

These contributions are elaborated in detail in Section 2.

2 Technical Overview

Leakage-resilient block ciphers: Defining a leakage-resilient block cipher is not straightforward. A black-box block cipher (without leakage) is expected to behave as a random permutation. However, since a random permutation is a theoretical primitive, it does not experience leakage. Besides, Berti *et al.* [9] discussed how deterministic constructions cannot provide indistinguishability-style notions in general. Luckily, Dodis and Pietrzak [27] presented a weakened notion for leakage-resilient PRPs. The queries are split into two categories, challenge and non-challenge queries. Non-challenge queries always use the real construction with leakage, while challenge queries use either the real construction without leakage or a random permutation, depending on which world the challenger is in. This notion captures the best scenario we can hope for without introducing unrealistic assumptions. We will refer to it as **STPRP ℓ 2** and it is the central security notion of the paper.

Generic reductions: We study the EtE framework using the **STPRP ℓ 2** assumption. Unlike the black-box setting, there is no unified AEAD security notion that captures both integrity and confidentiality in the presence of leakage. Bellizia *et al.* [8] showed a classification of leakage-resilient AEAD notions in the presence of leakage. For integrity, the strongest notion is Ciphertext Integrity with nonce Misuse and encryption and decryption Leakage (**CIML2**). A weaker notion is Ciphertext Integrity with nonce Misuse and encryption Leakage (**CIML1**). We show that **STPRP ℓ 2** with an appropriate encoding scheme tightly implies **CIML1**. We also show that **STPRP ℓ 2** implies security against reforgeability: we allow the challenger to tolerate r successful forgeries during which the adversary gets decryption leakage, then we disallow decryption leakage for any new forgery attempts. We call this notion r -**CIML2**.

On the other hand, the relation between **STPRP ℓ 2** and **CIML2** is more complicated. Because **STPRP ℓ 2** does not allow challenge leakage, and the leaky oracles are the same in both the real and ideal worlds, an **STPRP ℓ 2** adversary can only win by querying its leak-free oracles. However, any **CIML2** adversary is expecting exclusive access to leaky oracles. If we try to make a security reduction where the **STPRP ℓ 2** simulates the **CIML2** games honestly, it can never win its own game, as it will never query its challenge oracles. Thus, we design the reduction such that the **STPRP ℓ 2** adversary *cheats* by selecting one forgery attempt, and

feeding it to its challenge oracle. The C1ML2 adversary is undisturbed up to the point of cheating, and if that forgery attempt was successful, it would still be successful in this simulation. A successful forgery at this particular point allows the STPRP ℓ 2 adversary to win its own game. However, we cannot say anything about the behaviour of the C1ML2 adversary after this point, while everything that preceded this point does not really help the STPRP ℓ 2 adversary. The reduction hinges on the STPRP ℓ 2 adversary being able to guess which forgery attempt is actually successful. This leads to a linear drop in security from STPRP ℓ 2 to C1ML2, where the security drops with the number of forgery attempts.

We study the implication from STPRP ℓ 2 to confidentiality notions. Particularly, we study Chosen-Ciphertext Attacks with nonce-misuse resilience and challenge leakage (CCAmL2) and Chosen-Ciphertext Attacks with nonce-misuse resistance and no challenge leakage (CCAM ℓ 2). We show that STPRP ℓ 2 also implies CCAmL2 in terms of privacy, but it cannot imply CCAM ℓ 2. As we discussed in the case of C1ML2, an honest STPRP ℓ 2 adversary simulating such game can never win its own game. However, there is a fundamental difference between C1ML2 and CCAmL2 that makes the argument we used previously infeasible. Even if the STPRP ℓ 2 picks a random challenge query and uses its challenge oracle for it, then the CCAmL2 does not get the leakage its expecting and we cannot make any claims about its winning chances. That being said, the positive takeaway from this section is that STPRP ℓ 2 implies CCAmL2 + C1ML2 (with a drop in the latter). This is one of the strongest leakage-resilient AEAD combinations, and shows that EtE itself is leakage-resilient. What remains is to show the possibility of an EtE instance with a stronger combination: CCAmL2 + CCAM ℓ 2 + C1ML2 (with smaller or no security drop). While this cannot be done generically, we show that it can be done for specific schemes.

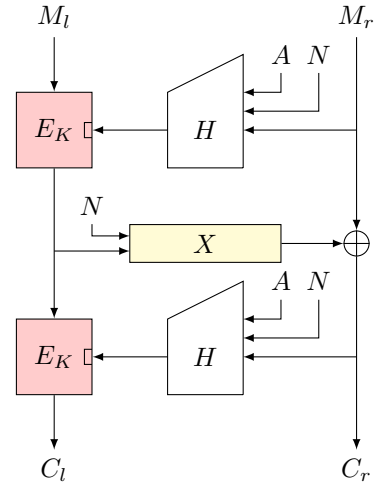


Fig. 1: The Motte-and-Bailey framework

Motte-and-Bailey: We define the Motte-and-Bailey (MaB) framework, depicted in Figure 1. The tweak in MaB consists of two parts, N , a fixed-length mandatory tweak, and A , a variable-length, potentially empty, tweak. This maps nicely to AEAD, where N can be used for the nonce and A can be used for the associated data. The framework consists of four main components: a leak-free TBC, a collision-resistant hash function, a leakage-resilient PRG, and an XOR addition.

The **MaB** framework draws inspiration from the **PIV** framework [64], with two major differences:

- **PIV** consists of two TBCs: one has a fixed block length and variable tweak length and is **STPRP**-secure, while the other has a fixed tweak length and variable block length and is only **STPRP**-secure against adversaries that never repeat tweaks. The secret key of this primitive requires protection against leakage. On the other hand, **MaB** does not have a secret key in the middle layer X , which is a completely public primitive. Informally, the middle layer of **MaB** can be seen as **eXtendable-Output Function (XOF)**.
- While **PIV** uses any variable-tweak-length TBC for its outer layers, **MaB** specifies these layers using a (public) collision-resistant hash function, and a fixed-tweak-length TBC.

The second difference can be seen as a special case of **PIV**. Indeed, if it was the only difference, then **MaB** could have been seen as an instance of **PIV**. The first difference is more subtle and deviates from **PIV**. In fact, the first difference qualitatively downgrades the security of **PIV**. It is not hard to see that in the black-box setting, the security proof of **PIV** can be adapted to **MaB**. This adaptation leads to security bound with complexity limits

$$D \cdot T < 2^n, D < 2^{n/2}$$

where D is the online data complexity and T is the offline number of queries to the primitive X . For **PIV**, only the second limit applies. In terms of leakage resilience, we argue that the new definition of X is an improvement. We could have built X as a keyed stream cipher. However, in order to maintain leakage resilience while keeping the cost low, we would have had to use a leakage resilient stream cipher. This would have added additional long-term secrets that need to be protected. Leakage-resilient stream ciphers that do not require full protection also suffer from similar security drops. Next, we show the **AEAD** security of **MaB** with **EtE**. For a specific encoding schemes achieves stronger **CIML2** security than what the generic reduction implies. This requires two ingredients:

- The encoding scheme is constructed such that the verification can be done before any part of the decrypted plaintext is computed, and if the verification fails, the decryption process is terminated.
- X behaves as a PRG even with challenge leakage.

The first ingredient is easy to achieve. The second ingredient is a stronger assumption than the one used for **STPRP** ℓ 2. However, in practice both assumption are achieved by similar constructions, which means that for most instances of **MaB**, we get this for free. With these two assumptions in addition to an extra feature of the hash function, we also prove **CCAmL2**. Particularly, in order to achieve **CCAmL2**, we need H to be secure in the **LORL2** notion of Berti *et al.* [9]. Informally, given any pair of messages (M_0, M_1) , the adversary cannot distinguish whether H absorbed M_0 or M_1 . Clearly, a public hash function with physical leakage cannot satisfy this assumption. Thus, not every instance of **MaB**

can satisfy this assumption. However, a candidate to satisfy this assumption is by applying a secret permutation to the nonce, such that the first input block to the hash function in each encryption query is a unique secret. This does not affect the earlier results, but allows us to prove **CCAmL2** security. Such a configuration achieves the strongest practically possible level of leakage resilience: **CCAmL2 + CCAM ℓ 2 + CIML2**.

*Instantiations: **Qaitbay** and **Alicante**:* In the remainder of the paper, we define two particular instances of **MaB**: **Qaitbay** and **Alicante**. The former is a realization of **MaB** based on the Sponge function and a tweakable block cipher, while the latter is a realization based on an ideal cipher. We also discuss how to instantiate both of them using existing primitives, what concrete security we get, and the limitations to the current instantiations. This shows the **MaB** is practical with current primitives, but it also opens the door for designing tailored primitives as future work.

Table 1 includes a summary of the parameters, primitives and security claims of each of the presented instantiations of **Qaitbay** and **Alicante**. The table shows significant advantage for **Motte-and-Bailey** compared to two prominent schemes. In the case of **HCTR2**, we considered two instantiations, one from standard AES, and another from **Rijndael-256-128** [22] (we shall say **Rijndael-n-k** for n -bit block and k -bit key). The structure of **HCTR2** does not follow that of **MaB**, so we mapped the primitives as closely as possible to the right column. We did not count the overhead from the underlying almost-XOR-universal hash function, represented in the table by $\star$. In the case of **Kravatte-WBC** [12], and to be fair, we have reduced the number of calls by half, since **Kravatte-WBC** uses half the number of rounds per call compared to **TurboSHAKE** [13]. However, any gain from this is offset by how **Kravatte-WBC** needs all the permutation calls to be heavily protected. We also need to note that while **Alicante** and **HCTR2** (with AES) have the same bit security, **Alicante** suffers from a time-data trade-off, while **HCTR2** does not. All the security numbers are without leakage. Since none of the existing Accordion modes is leakage-resilient, we also included **TEDT** [9], a leakage-resilient AEAD, as a point of reference. Compared to **TEDT**, a leakage-resilient AEAD from ideal ciphers, the overhead of **Alicante-1** is only one extra hash call, while for **Alicante-2** the overhead also includes one extra protected call. These overheads are expected as **Alicante** targets a significantly stronger security model. Compared to **HCTR2**, the cost of **Alicante** is roughly double, in terms of pure number of calls. However, **HCTR2** requires strong protection (leak-free assumptions) for every BC and hash call, while for **Alicante**, the number of leak-free calls is either 2 or 3, making **Alicante** significantly cheaper for protected implementations.

3 Preliminaries: Generic Security Notions

Notation We denote sets by using calligraphic letters: $\mathcal{A}$. Additionally, we will denote the cardinality of said sets by $|\mathcal{A}|$. Moreover, if $\mathcal{A}$ is a finite set, we write

Table 1: Summary of the plausible instantiations and their security. # Calls refers to the number of unprotected/lightly protected permutation or TBC calls, where l is the bit-length of M .

Scheme	TBC	H/X	CCAmL2	Sec.	Key Rec.	Key	Prot. Calls	# Calls
Qaitbay-1	XPX[PHOTON-256]	TurboSHAKE-256	×	128	128	256	4	$3 \lceil \frac{(l-256)}{1344} \rceil$
Qaitbay-2	XPX[PHOTON-256]	TurboSHAKE-256	✓	128	128	256	5	$3 \lceil \frac{(l-256)}{1344} \rceil$
Qaitbay-1	Rijndael-IC	TurboSHAKE-256	×	128	128	256	2	$3 \lceil \frac{(l-256)}{1344} \rceil$
Qaitbay-2	Rijndael-IC	TurboSHAKE-256	✓	128	128	256	3	$3 \lceil \frac{(l-256)}{1344} \rceil$
Qaitbay-1	Rijndael-PRP	TurboSHAKE-256	×	128	128	128	4	$3 \lceil \frac{(l-256)}{1344} \rceil$
Qaitbay-2	Rijndael-PRP	TurboSHAKE-256	✓	128	128	128	5	$3 \lceil \frac{(l-256)}{1344} \rceil$
Alicante-1	Deoxys-128-640	Deoxys-128-640	×	64	128	128	2	$2 \lceil \frac{(l-256)}{512} \rceil + \lceil \frac{(l-256)}{64} \rceil$
Alicante-2	Deoxys-128-640	Deoxys-128-640	✓	64	128	128	3	$2 \lceil \frac{(l-256)}{512} \rceil + \lceil \frac{(l-256)}{64} \rceil$
HCTR2	AES	AES/UHF	×	64	—	256	$\lceil \frac{l}{128} \rceil + \star$	0
HCTR2	Rijndael	Rijndael/UHF	×	128	—	> 256	$\lceil \frac{l}{256} \rceil + \star$	0
Kravatte-WBC	Keccak-p[6]	Keccak-p[6]	×	128	128	128	$3 \lceil \frac{l}{800} \rceil$	0
TEDT	Deoxys-128-640	Deoxys-128-640	✓	64	128	128	2	$\lceil \frac{l}{512} \rceil + \lceil \frac{l}{64} \rceil$

$x \xleftarrow{\$} \mathcal{S}$ to denote that x is sampled uniformly at random from $\mathcal{S}$. We use $\emptyset$ to denote the empty set, and, abusing notation, we also use $\emptyset$ to denote the empty string. Given an algorithm $\mathcal{A}$, we denote $\mathcal{A}$ outputting a value x by $\mathcal{A} \Rightarrow x$ or $x \leftarrow \mathcal{A}$. Additionally, when defining a new variable we will use the symbol $\stackrel{\text{def}}{=}$. Let $n \in \mathbb{N}$ such that $n \geq 1$. Let We will abbreviate the set $\{1, \dots, n\}$ as $[n]$. Let X be a variable-length bit string, then $|X|$ is its bit length, represented as a 64-bit string. For $n \in \mathbb{N}$, $\xleftarrow{n}$ splits an arbitrary-length bit string into n -bit blocks, with the last block potentially shorter than n bits, and $\xleftarrow{n,*}$ splits an arbitrary-length bit string of length at least n bits into an n -bit block and an arbitrary-length bit string that contains the remainder of the bits. Finally, we denote by $\text{tperm}(\mathcal{T}, \mathcal{M})$ the set is a tweakable uniformly random permutations, that is, functions from $\mathcal{T} \times \mathcal{M} \mapsto \mathcal{M}$, so that, for any $t \in \mathcal{T}$, $R(t, \cdot)$ is an independent uniformly random permutation.

In our constructions, we use a single-input collision-resistant hash function to absorb three different string. For that, we require an injective padding function ipad . For the case of MaB, we leave the exact definition of this function to the application, under the condition that it is impossible to find two sets of inputs that lead to the same string after padding, i.e., the padding must be injective. An example of such function is defined in [33]. Nevertheless, for our constructions, Qaitbay and Alicante, padding is explicitly defined and used when needed (cf. Algorithms 2 to 4).

Collision Resistance of Hash Functions: Let $H : \mathcal{K}_h \times \mathcal{M} \rightarrow \mathcal{X}$ be a hash function. A hash function H is called (ϵ_{cr}, t) -collision-resistant if for every t -bounded adversary $\mathcal{A}$ (i.e. running in time at most t), the probability that $\mathcal{A}(s)$ outputs a pair of distinct inputs $(M_1, M_2) \in \mathcal{M}^2$, such that $H_s(M_1) = H_s(M_2)$ and $M_1 \neq M_2$, is bounded by ϵ_{cr} , with $s \xleftarrow{\$} \mathcal{K}_h$ picked uniformly at random:

$$\Pr[s \xleftarrow{\$} \mathcal{K}_h, (M_1, M_2) \leftarrow \mathcal{A}(s) \text{ s.t. } M_1 \neq M_2, H_s(M_1) = H_s(M_2)] \leq \epsilon_{cr}.$$

The hash key is revealed to the adversary at the beginning of the security game. We omit the hash key from the notation when understood from context. We emphasize that the hash key is public, and in practice can be represented by a simple fixed-length constant IV . This is a standard definition for collision resistance of public hash functions and has been particularly useful in leakage-resilient constructions [11].

Authenticated Encryption with Associated Data (AEAD): An AEAD scheme is a pair of functions $\Pi = (E, D)$. $E : \mathcal{K} \times \mathcal{N} \times \mathcal{A} \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ is the encryption function that takes secret key $K \in \mathcal{K}$, nonce $N \in \mathcal{N}$, associated data $A \in \mathcal{A}$ and plaintext $M \in \{0, 1\}^*$ and returns ciphertext $C \in \{0, 1\}^*$. $D : \mathcal{K} \times \mathcal{N} \times \mathcal{A} \times \{0, 1\}^* \rightarrow \{0, 1\}^* \cup \{\perp\}$ is the decryption function that takes secret key $K \in \mathcal{K}$, nonce $N \in \mathcal{N}$, associated data $A \in \mathcal{A}$ and ciphertext $C \in \{0, 1\}^*$ and returns plaintext $M \in \{0, 1\}^*$ or the symbol $\perp$.

3.1 Security Notions for Leakage Resilience

In this section, we summarize the security notions we use in our study. Most of these notions already exist in the literature, and we sometimes rename them to match our notation, without changing their nature. The notation is mostly inspired by Bellizia *et al.* [8]:

- ℓ refers to notions with no challenge leakage, while L refers to notions with challenge leakage.
- For notions with a nonce, m refers to misuse resilience, M refers to misuse resistance, and the lack of either refers to nonce-respecting notions.
- A suffix 1 refers to the presence of leakage in only the first oracle, while 2 refers to the presence of leakage in two oracles.

An informal description of the leakage functions is given in Appendix A.

Strong Tweakable Pseudo-Random Permutation without challenge leakage (STPRP ℓ 2): This notion was introduced by Dodis and Pietrzak [27] and a non-adaptive variant was introduced by Faust *et al.* [30]. Let $\tilde{\Pi} : \mathcal{K} \times \mathcal{T}_w \times \mathcal{M} \rightarrow \mathcal{M}$ be a family of keyed length-preserving permutations with key space $\mathcal{K}$ and tweak space $\mathcal{T}_w$. For each pair $(K, T) \in \mathcal{K} \times \mathcal{T}_w$, $\tilde{\Pi}_K^T$ is a length-preserving permutation over the message space $\mathcal{M}$. Let L be the associated leakage function. $\tilde{\Pi}_K^T$ is the leaky oracle of $\tilde{\Pi}_K^T$. Let $\tilde{R} : \mathcal{T}_w \times \mathcal{M} \rightarrow \mathcal{M}$ be a length-preserving Tweakable Uniformly Random Permutation (TURP), where for each $T \in \mathcal{T}_w$ and each message length l , $\tilde{R}^T$ is a permutation sampled uniformly at random from the set of all length-preserving permutations over $\mathcal{M}$. Let $\mathcal{A}$ be an adversary with access to four oracles that returns a single bit. The STPRP ℓ 2 advantage of $\mathcal{A}$ is defined as

$$\text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}) \stackrel{\text{def}}{=} \left| \Pr[K \xleftarrow{\$} \mathcal{K} : \mathcal{A}^{W_L, W_R} \Rightarrow 1] - \Pr[K \xleftarrow{\$} \mathcal{K}, \tilde{R} \xleftarrow{\$} \mathcal{R} : \mathcal{A}^{W_L, W_I} \Rightarrow 1] \right|.$$

where $W_L = (\tilde{\Pi}_L, \tilde{\Pi}_L^{-1})$, $W_I = (\tilde{R}, \tilde{R}^{-1})$ and $W_R = (\tilde{\Pi}, \tilde{\Pi}^{-1})$ and $\mathcal{R}$ is the set of all length-preserving tweakable permutations of the same tweak and plaintext spaces. We say $\tilde{\Pi}$ is STPRP ℓ 2-secure if the STPRP ℓ 2 advantage of any adversary is negligible. More concretely, we say that $\tilde{\Pi}$ is $(q_e, q_d, q_{ce}, q_{cd}, t)$ -STPRP ℓ 2-secure, if the advantage is negligible for any adversary that make at most q_e and q_d leaky encryption and decryption queries ins total and at most q_{ce} and q_{cd} challenge encryption and decryption queries and runs in time at most t . $\tilde{\Pi}$ maybe defined with respect to one or more ideal primitives. In that case, the adversary is granted access to these primitives and the number of queries made to each primitive is part of the adversarial resources.

Ciphertext Integrity with Leakage (CIML2): In a CIML2 game the goal of the adversary is to produce a valid fresh ciphertext, while given access to access to leakage both in encryption and decryption. The adversary can repeat nonces even in challenge queries. Formally, we define the advantage of an adversary, $\mathcal{A}$ as in this game as

$$\text{Adv}_{\Pi}^{\text{CIML2}}(\mathcal{A}) \stackrel{\text{def}}{=} \Pr \left[\mathcal{A}^{\Pi_E^L, \Pi_D^L} \Rightarrow (N, T, M, C) : \Pi_E(N, T, M) = C \right].$$

Chosen Ciphertext Attack with Leakage (CCA). At the beginning of the game a random $b \xleftarrow{\$} \{0, 1\}$ is sampled, the goal of $\mathcal{A}$ is to guess b . During a challenge query, the adversary, $\mathcal{A}$ queries two plaintexts (M_0, M_1) and receives the encryption of M_b , i.e., C_b . That is, it a so-called *left-or-right* game.

However, CCAML2 is widely believed to be impossible [8], except in trivial settings. Hence, we must find more realistic alternatives [4, 37]. In particular, the notions of CCAML2 and CCAML2 are achievable. This notions are quite similar, except in CCAML2, nonce misuse is allowed everywhere, but there is no leakage during challenge queries, while in CCAML2, there is leakage everywhere, but challenge queries may only use fresh nonces. The privacy game is defined as follows:

1. At the beginning of the game, the challenger selects a random bit b .
2. For every non-challenge query made by the adversary, the challenger queries either E or D, and returns the answer with the corresponding leakage.
3. For each challenge query, the adversary sends N, A, M_0, M_1 and the challenger returns $E(N, A, M_b)$ without leakage.

In the CCAML2 game, the adversary can repeat nonces in all queries. In the CCAML2 game, the adversary is required to use fresh, non-repeating nonces for challenge queries, but it can query a special challenge leakage oracle with $(N, A, E(N, A, M_b))$. This oracle returns the corresponding leakage but not the value of M_b . At the end of the game, the adversary returns either 0 or 1. Formally, we define the advantage of an adversary, $\mathcal{A}$ as in these games as

$$\text{Adv}_{\Pi}^{\text{CCAML2/CCAML2}}(\mathcal{A}) \stackrel{\text{def}}{=} |\Pr[\mathcal{A} \Rightarrow 1 | b = 1] - \Pr[\mathcal{A} \Rightarrow 1 | b = 0]|.$$

Leakage-Resilient Pseudo-Random Generator (PRG ℓ 1, PRGL1, and GuessL1): Let $X : \mathcal{S} \times \mathcal{N} \times \mathbb{Z}^+ \rightarrow \{0, 1\}^*$ be a function that takes a random seed $S \in \mathcal{S}$, a public parameter $N \in \mathcal{N}$ and a positive integer l and returns a bit-string of length l . The PRG ℓ 1 game is defined as follows:

1. The challenger samples a random bit $w \xleftarrow{\$} \{0, 1\}$ at the beginning of the game.
2. An adversary $\mathcal{A}$ makes a query $(N, l, b) \in \mathcal{N} \times \mathbb{Z}^+ \times \{0, 1\}$.
3. The challenger samples a uniformly random seed $S \xleftarrow{\$} \mathcal{S}$.
4. If $b = 0$, the challenger returns $X_L(S, N, l)$.
5. If $b = 1$ and $w = 0$, the challenger returns $X(S, N, l)$.
6. Otherwise, the challenger samples a uniformly random string $Z \xleftarrow{\$} \{0, 1\}^l$ and returns Z .

The adversary can also make q_p queries to X with a chosen seed, we call this oracle X_P . The adversary makes q_l queries with $b = 0$ and q_e queries with $b = 1$. The PRG ℓ 1 advantage of $\mathcal{A}$ is defined as

$$\text{Adv}_X^{\text{PRG}\ell 1}(\mathcal{A}) \stackrel{\text{def}}{=} |\Pr[\mathcal{A}^{X_P, X_L, X} \Rightarrow 1] - \Pr[\mathcal{A}^{X_P, X_L, \$} \Rightarrow 1]|,$$

where $\$$ is the oracle that returns uniformly random bit strings. The leakage-resistant variant of this notion is defined by the interaction of the adversary with (X_P, X_L) or $(X_P, \$_L)$, where $\$$ outputs random strings but the correct leakage of X . The PRGL1 advantage of $\mathcal{A}$ is defined as

$$\text{Adv}_X^{\text{PRGL1}}(\mathcal{A}) \stackrel{\text{def}}{=} |\Pr[\mathcal{A}^{X_P, X_L} \Rightarrow 1] - \Pr[\mathcal{A}^{X_P, \$_L} \Rightarrow 1]|,$$

The adversary makes q_e queries challenge queries. The adversary can also make q_p queries to X with a chosen seed. Clearly, PRGL1 implies PRG ℓ 1, as the adversary can simply ignore the leakage from some queries. A PRGL1 adversary can simulate a PRG ℓ 1 game using only primitive queries for the PRG ℓ 1 leakage queries and use its second oracle for the challenge queries, ignoring the leakages. We also define the GuessL1 game as follows:

1. An adversary $\mathcal{B}$ makes a query $(N, l, 0) \in \mathcal{N} \times \mathbb{Z}^+ \times \{0, 1\}$.
2. The challenger samples a uniformly random seed $S \xleftarrow{\$} \mathcal{S}$.
3. The challenger returns $X_L(S, N, l)$.

The adversary can also make q_p queries to X with a chosen seed, we call this oracle X_P . At the end of the games, $\mathcal{B}$ returns a set $\mathcal{G}$ of size q_g . $\mathcal{B}$ wins if $S \in \mathcal{G}$. We define $\mathcal{B}$'s advantage as

$$\text{Adv}_X^{\text{GuessL1}[q_g]}(\mathcal{B}) \stackrel{\text{def}}{=} \Pr[\mathcal{G} \leftarrow \mathcal{B}^{X_P, X_L} : S \in \mathcal{G}].$$

We note that this notion can be trivially broken with a brute-force attack, where $\mathcal{B}$ observes a collision between S and a query to X_P . This occurs with probability about $q_p/|\mathcal{S}|$. In the presence of leakage, this probability can be improved.

Key Unpredictability with two traces (2-up): Berti *et al.* [10] introduced the 2-up security notion as a notion for security against 2-trace attacks on ideal ciphers. GuessL1 can be seen as a generalization of 2-up. The game work almost identically to the GuessL1 game except the adversary is given two queries to the ideal cipher with the same secret key S . Then, the adversary wins if it can guess the key. Against this game, the advantage of the adversary is defined as

$$\text{Adv}_X^{2\text{-up}[qg]}(\mathcal{B}) \stackrel{\text{def}}{=} \Pr[\mathcal{G} \leftarrow \mathcal{B}^{\text{IC}_P, \text{IC}_L} : S \in \mathcal{G}].$$

3.2 EtE: Encode-then-Encipher

The Encode-then-Encipher (EtE) is an AEAD construction that was introduced by Rogaway and Bellare [6]. It is a tagless AEAD construction, but rather integrity is provided by the encoding scheme. Meanwhile, confidentiality is provided by the enciphering mechanism. We limit our analysis to sparse encoding schemes.

Sparse Encoding. Let $S = (\text{encode} : \{0, 1\}^* \rightarrow \{0, 1\}^*, \text{decode} : \{0, 1\}^* \rightarrow \{0, 1\}^* \cup \{\perp\})$ be an encoding scheme, such that $\text{decode}(\text{encode}(M)) = M$, and let δ be a real number. We say that S is δ -dense if for all $n \in \mathbb{N}$, it holds that

$$\Pr[\bar{M} \xleftarrow{\$} \{0, 1\}^n : \text{decode}(\bar{M}) \in \{0, 1\}^*] \leq \delta. \quad (1)$$

In other words, for every possible codeword length, only a δ -fraction of all codewords of that length are valid codewords, i.e., decode to valid messages, while the rest decode to $\perp$ (invalid). The most useful and natural example of a δ -dense encoding scheme is the following:

$$\text{encode}(M) \Rightarrow 0^t \parallel M; \quad \text{decode}(C) \Rightarrow \begin{cases} M & \text{if } C = 0^t \parallel M, \\ \perp & \text{otherwise,} \end{cases}$$

which is clearly a δ -dense encoding scheme with $\delta = 2^{-t}$.

The EtE construction consists of an encoding scheme $S = (\text{encode}, \text{decode})$ and an enciphering (TWBC) scheme $\tilde{\Pi}$, where $\tilde{\Pi}.\mathcal{E}$ is enciphering algorithm and $\tilde{\Pi}.\mathcal{D}$ is the corresponding inverse. Conceptually, a enciphering scheme $\Pi = (\Pi.\mathcal{E}, \Pi.\mathcal{D})$ is a variable-input-length TBC [51], which takes a key K , a tweak T , and a plaintext M as input, and outputs a ciphertext C . $\tilde{\Pi}$ is typically assumed to be STPRP-secure. In EtE, the nonce and associated data can be consumed as the tweak. We remark that although EtE takes a nonce, it is in fact nonce misuse resistant. Since its introduction, the EtE construction has been a valuable tool in building robust and misuse-resistant AEAD.

4 Generic Relations: Building EtE from STPRP ℓ 2

In this section, we elevate EtE to the realm of leakage resilience. Instead of assuming $\tilde{\Pi}$ is STPRP-secure, we now assume it is STPRP ℓ 2-secure. Then, we provide security reductions for EtE, from STPRP ℓ 2 security to CIML1, CIML2, and CCAM ℓ 2.

Theorem 1. *Let $\mathcal{B}$ be a CIML1 adversary against EtE with a δ -dense coding scheme and a TBC $\tilde{\Pi}$, that makes q queries, where $q = q_e + q_f$, such that q_e is the number of encryption queries and q_f is the number of forgery queries. If $\mathcal{B}$ produces a successful forgery with non-negligible probability, then there exists an STPRP ℓ 2 adversary $\mathcal{A}$ against $\tilde{\Pi}$ that makes $q = q_e + q_f$ queries, where q_e is the number non-challenge queries and q_f is the number of challenge queries, such that*

$$\text{Adv}_{\text{EtE}[\delta, \tilde{\Pi}]}^{\text{CIML1}}(\mathcal{B}) \leq \text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}) + 2q_f\delta.$$

Theorem 2. *Let $r \in \mathbb{Z}^+$. If $\mathcal{B}$ is a r -CIML2 adversary against EtE with a δ -dense coding scheme and a TBC $\tilde{\Pi}$, that makes q queries, where $q = q_e + q_f^a + q_f^b$, such that q_e is the number of encryption queries and q_f^a is the number of forgery queries after the r^{th} successful forgery, while the number of forgery queries until (and including) the r^{th} successful forgery is q_f^b , and produces a successful forgery after the r^{th} successful forgery with non-negligible probability. Then, there exists an STPRP ℓ 2 adversary $\mathcal{A}$ against $\tilde{\Pi}$ that makes $q = q_e + q_f^b + q_f^a$ queries, where $q_e + q_f^b$ is the number non-challenge queries and q_f^a is the number of challenge queries, such that*

$$\text{Adv}_{\text{EtE}[\delta, \tilde{\Pi}]}^{r\text{-CIML2}}(\mathcal{B}) \leq \text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}) + 2q_f^a\delta.$$

The proof of Theorem 1 is deferred to Appendix B.1. It is a simple reduction where all the encryption queries are non-challenge queries, and all the forgery attempts are simulated using challenge queries. If the forgery is successful, then it is the real world (the probability of successful forgery in the ideal world is negligible if δ is negligible.) The proof of Theorem 2 is almost the same as the proof of Theorem 1, except $\mathcal{A}$ uses the non-challenge leaky decryption oracle at first, and keeps track of how many successful forgeries have been made. Once r successful forgeries have occurred, $\mathcal{A}$ switches to the challenge oracle. Furthermore, it is easy to see that 0-CIML2 is equivalent to CIML1. However, since there is no r for which r -CIML2 is equivalent to CIML2, and due to the fact that r has to be non-adaptively chosen, CIML2 is not implied by this theorem and requires further analysis. Clearly, if the number of successful forgeries is less than r , the adversary cannot win and the claim is trivial. However, Theorems 1 and 2 give strong evidence that STPRP ℓ 2 offers strong integrity guarantees for ciphertext integrity in a generic sense. Theorem 3 shows that STPRP ℓ 2 implies CIML2 with a linear gap. We do not have a matching attack that exploits this linear gap, and it may be an artefact of the proof or a limitation of the model. Later, we will show that the gap between CIML2 and STPRP ℓ 2 can be closed for concrete instantiations (with additional assumptions). We also note that r -CIML2 is just a weaker version of the *challenge-then-leak* philosophy.

Theorem 3. *Let $\mathcal{B}$ be a CIML2 adversary that wins with non-negligible probability after making q_e encryption queries and q_f forgery attempts, and runs in time t . Then, there exists an STPRP ℓ 2 adversary $\mathcal{A}$ that makes at most q_e encryption*

queries and at most q_f decryption queries and runs in time $O(t)$, such that

$$\text{Adv}_{\text{EtE}[\delta, \tilde{\Pi}]}^{\text{CIML2}}(\mathcal{B}) \leq q_f \text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}) + q_f \delta.$$

Proof. For $i \in \mathbb{Z}^+$, let $\mathcal{A}_i$ be an STPRP $\ell 2$ adversary that works as follows:

1. $\mathcal{A}_i$ runs $\mathcal{B}$.
2. As long as $\mathcal{B}$ has made less than i forgery attempts, $\mathcal{A}_i$ simulates the CIML2 game using the encoding scheme and its leaky oracles.
3. At the i^{th} forgery attempt, $\mathcal{A}_i$ sends the forgery to its challenge leak-free decryption oracle.
4. $\mathcal{A}_i$ decodes the response.
5. If the forgery is successful, $\mathcal{A}_i \Rightarrow 1$. Otherwise, $\mathcal{A}_i \Rightarrow 0$.
6. $\mathcal{A}_i$ terminates the game.

First, we consider the ideal world. Consider that the challenge oracle is simulated using lazy sampling. $\mathcal{A}_i$ queries the challenge oracles once and gets a response sampled from a uniform distribution. Thus, the probability that the response is a valid codeword is exactly δ . In other words,

$$\Pr[\mathcal{A}_i^{\tilde{\Pi}_L, \tilde{\Pi}_L^{-1}, \tilde{R}, \tilde{R}^{-1}} \Rightarrow 1] = \delta.$$

Next, we analyze the real world. It is clear that if $\mathcal{B}$ wins, then $\exists i$, s.t. $1 \leq i \leq q_f$ and $\mathcal{A}_i \Rightarrow 1$. Thus,

$$\sum_{i=1}^{q_f} \Pr[\mathcal{A}_i^{\tilde{\Pi}_L, \tilde{\Pi}_L^{-1}, \tilde{\Pi}, \tilde{\Pi}^{-1}} \Rightarrow 1] \geq \Pr[\mathcal{B} \text{ wins}] = \text{Adv}_{\text{EtE}[\delta, \tilde{\Pi}]}^{\text{CIML2}}(\mathcal{B}).$$

Combining the two bounds, we get,

$$\begin{aligned} \sum_{i=1}^{q_f} \text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}_i) &= \sum_{i=1}^{q_f} \Pr[\mathcal{A}_i^{\tilde{\Pi}_L, \tilde{\Pi}_L^{-1}, \tilde{\Pi}, \tilde{\Pi}^{-1}} \Rightarrow 1] - \Pr[\mathcal{A}_i^{\tilde{\Pi}_L, \tilde{\Pi}_L^{-1}, \tilde{R}, \tilde{R}^{-1}} \Rightarrow 1] \geq \\ &\quad \text{Adv}_{\text{EtE}[\delta, \tilde{\Pi}]}^{\text{CIML2}}(\mathcal{B}) - q_f \delta. \end{aligned}$$

Let $\mathcal{A} = \mathcal{A}_i$ such that $\text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}_i) = \max_{1 \leq j \leq q_f} \text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}_j)$. Then,

$$q_f \text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}) \geq \sum_{i=1}^{q_f} \text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}_i) \geq \text{Adv}_{\text{EtE}[\delta, \tilde{\Pi}]}^{\text{CIML2}}(\mathcal{B}) - q_f \delta.$$

Thus,

$$\text{Adv}_{\text{EtE}[\delta, \tilde{\Pi}]}^{\text{CIML2}}(\mathcal{B}) \leq q_f \text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}) + q_f \delta.$$

□

Next, we show the confidentiality of EtE with leakage.

Theorem 4. *If $\mathcal{B}$ is a $\text{CCAM}\ell 2$ adversary against EtE . Then, there exists an $\text{STPRP}\ell 2$ adversary $\mathcal{A}$ against $\tilde{\Pi}$, such that*

$$\text{Adv}_{\text{EtE}[\delta, \tilde{\Pi}]}^{\text{CCAM}\ell 2}(\mathcal{B}) \leq \text{Adv}_{\tilde{\Pi}}^{\text{STPRP}\ell 2}(\mathcal{A}) + \frac{q(q-1)}{2^{\min(|M|)+1}}.$$

and both adversaries make the same number of queries.

The proof of Theorem 4 is deferred to Appendix B.2. It seems rather intuitive that $\text{STPRP}\ell 2$ cannot imply AEAD notions that involve challenge leakage, such as CCAmL2 , or forgery with leakage such as CIML2 (without a significant gap in the reduction). An intuitive way to argue this separation is that during any reduction, the adversary against $\text{STPRP}\ell 2$ needs to simulate these games, all of which require forwarding the challenges against the AEAD games to the leaky $\text{STPRP}\ell 2$ oracles. However, these queries do not help the $\text{STPRP}\ell 2$ adversary as they are always made to the real-world leaky oracles, while the adversary can win only by making challenge queries to the leak-free oracles. Of course, these queries still help the adversary, but the adversary still needs to devise a procedure to generate its challenge oracles. The gap between CCAmL2 and $\text{STPRP}\ell 2$ is not artificial, as we can show that MaB can be instantiated such that it is $\text{STPRP}\ell 2$ -secure but not CCAmL2 -secure. We discuss this in details in Section 6, where we also give a strengthened version that achieves CCAmL2 security. We also note these results show that in a generic way, a leakage-resilient Accordion mode still provides leakage-resilience as an AEAD, albeit not optimal. For confidentiality, it is considered impossible to achieve CCAML2 , making the best achievable combinations are either $\text{CCAM}\ell 2$ or CCAmL2 , and we prove that the former is generically achievable from leakage-resilient Accordion modes.

5 The Motte-and-Bailey Framework

In this section, we introduce the **Motte-and-Bailey** framework. MaB aims to provide high security guarantees, even in the presence of leakage. It achieves this through a layered construction, using two fully protected, fixed size block ciphers and a less protected PRG, but the remainder remains unprotected. A detailed description can be found in Algorithm 1. Furthermore, we analyze its security and state the main security claims related to this construction, which can be found below.

Theorem 5. *Let H be an (ϵ_{CR}, t') -collision-resistant hash function with unbounded leakage, $\tilde{E}$ is a leak-free TBC, and X is a leakage-resilient random function, where $\tilde{E}$ and H are sampled independently of X . Then, for any adversary $\mathcal{A}$ against the $\text{STPRP}\ell 2$ game makes at most:*

$$\begin{aligned} q_e: & \text{Encryption queries,} & q_d: & \text{Decryption queries,} \\ q_{ce}: & \text{Challenge encryption queries,} & q_{cd}: & \text{Challenge decryption queries,} \\ q_p: & \text{Primitives (offline) queries.} \end{aligned}$$

where furthermore we denote $q = q_e + q_{ce} + q_d + q_{cd}$ and $Q = q + q_p$. Then we have:

$$\begin{aligned} \text{Adv}_{\text{MaB}[\tilde{E}, H, X]}^{\text{STPRP}\ell 2}(\mathcal{A}) &\leq 2\text{Adv}_{\tilde{H}}^{\text{STPRP}}(\mathcal{A}_1) \\ &\quad + \text{Adv}_X^{\text{PRG}\ell 1}(\mathcal{A}_3) + (q_e + q_d)\text{Adv}^{\text{GuessL1}[2q+q_p]}(\mathcal{A}_2) \\ &\quad + \epsilon_{\text{CR}} + \frac{17q^2}{2^n} + \frac{(2(q_e + q_d))^2}{2^n} + \frac{(q_e + q_d)}{2^n}, \end{aligned}$$

where $\mathcal{A}_1$, $\mathcal{A}_2$ and $\mathcal{A}_3$ are any adversaries in their corresponding games. $t' > t + 2qt_H$, where t_H is the time of one call to H .

Proof. We build a sequence of hybrid games. Let E_i be the event that the adversary returns 1 in game G_i .

G0: This game represents the real-world, that is, the adversary has access to the construction, with real leakage. The construction is described in Algorithm 1(left).

G1: We replace the leak free TBC $\tilde{E}_K$ with a TURP. There exists an adversary $\mathcal{A}_1$, such that

$$|\Pr[E_0] - \Pr[E_1]| \leq \text{Adv}_{\tilde{E}}^{\text{STPRP}}(\mathcal{A}_1).$$

G2: We consider an adversary $\mathcal{A}_2$ that has access to the construction, and forwards $\mathcal{A}$'s queries to the construction and sends back the responses to $\mathcal{A}$. For every query, $\mathcal{A}_2$ makes two queries to the hash function H : $H(N\|A\|A\|M_r)$ and $H(N\|A\|A\|C_r)$ and adds the input-output pair to a hash table $\mathcal{H}$, initialized as empty. At the end of the game, $\mathcal{A}_2$ checks $\mathcal{H}$ and terminates the game if it includes a collision. From the assumption on H ,

$$|\Pr[E_1] - \Pr[E_2]| \leq \epsilon_{\text{CR}}.$$

where H is be $(t', \epsilon_{\text{CR}})$ -collision-resistant, such that $t' > t + 2qt_H$, where t_H is the time of one call to H . In the remainder of the proof, we set T_w to be the input to H for each hash query.

G3: We replace the composition of h and the TURP with a function $\tilde{F}$ that operates as follows: For each forward query (T_w, Y) (resp. backward query (T_w, Z)), it checks if there is a previous query such that $\tilde{F}(T_w, Y) = Z$, in which

Algorithm 1 Plain Motte-and-Bailey (left) and strengthened MaB (right). Protected TBC calls are colored in teal.

1: $\alpha \leftarrow \mathbf{H}(N\ A\ A\ M_r)$	1: $\gamma \leftarrow \tilde{E}_K^{1,0}(N)$
2: $\theta \leftarrow \tilde{E}_K^\alpha(M_l)$	2: $\alpha \leftarrow \mathbf{H}(\gamma\ A\ A\ M_r)$
3: $C_r \leftarrow M_r \oplus \mathbf{X}(N\ \theta)$	3: $\theta \leftarrow \tilde{E}_K^{0,\alpha}(M_l)$
4: $\alpha \leftarrow \mathbf{H}(N\ A\ A\ C_r)$	4: $C_r \leftarrow M_r \oplus \mathbf{X}(N\ \theta)$
5: $C_l \leftarrow \tilde{E}_K^\alpha(\theta)$	5: $\alpha \leftarrow \mathbf{H}(\gamma\ A\ A\ C_r)$
	6: $C_l \leftarrow \tilde{E}_K^{0,\alpha}(\theta)$

case it returns Z (resp. Y). Otherwise, it returns a uniformly random values. This transition is covered by the PRP-PRF-switch-like result of [40, Lemma 1], with

$$|\Pr[E_2] - \Pr[E_3]| \leq \frac{4q^2}{2^n}.$$

G4: We terminate the game if one of the following occurs: There exists two queries to $\tilde{F}$ such that

- $\tilde{F}(T_w, Y) = \tilde{F}(T'_w, Y')$ such that $(T_w, Y) \neq (T'_w, Y')$, or
- $\tilde{F}^{-1}(T_w, Z) = \tilde{F}^{-1}(T'_w, Z')$ such that $(T_w, Z) \neq (T'_w, Z')$.

The probability of this event is at most $\binom{2q}{2}/2^n$. Thus,

$$|\Pr[E_3] - \Pr[E_4]| \leq \frac{\binom{2q}{2}}{2^n} \leq \frac{4q^2}{2^n}.$$

We note that this transition is needed although some of the collisions are already covered in the previous transition. This is because the previous transition only deals with collisions of the form $\tilde{F}(T_w, Y) = \tilde{F}(T_w, Y')$ or $\tilde{F}^{-1}(T_w, Z) = \tilde{F}^{-1}(T_w, Z')$, but does not deal with collisions that involve different tweaks.

G5: This game is constructed such that the inputs to X are: (a) uniformly distributed, and (b) collision free.

$$(N_i, T_i, M_{r_i} \parallel M_{l_i}, C_{r_i} \parallel C_{l_i}, \theta_i).$$

be the tuple of the i^{th} non-challenge query, augmented with the corresponding θ_i .

- B5_F: there exists an encryption (challenge or non-challenge) query (N, A, M_l, M_r) where

$$(N_i, A_i, \theta_i, C_{r_i}) = (N, A, M_l, M_r).$$

- B5_B: there exists a decryption (challenge or non-challenge) query (N, A, M_l, M_r) where

$$(N_i, A_i, \theta_i, M_{r_i}) = (N, A, C_l, C_r).$$

- B5_P: there exists an offline query (χ_j, ι_j) such that:

$$\chi_j = (N_i, \theta_i).$$

Additionally, we denote $B5 = B5_F \vee B5_B \vee B5_P$. Intuitively, if an adversary can make some query, where it knows the intermediate value θ , then it can compute the right value, C_r (respectively M_r). In the real world, $C_r = X(\theta_i) \oplus M_r$ (respectively vice-versa), while C_r is sampled randomly in the ideal world, which makes distinguishing trivial. Hence, the game terminates in any of these events occurs.

If either of these events occur, then there is an adversary $\mathcal{A}_2$ against the GuessL1 security of X . $\mathcal{A}_1$ has offline access to X , runs $\mathcal{A}$, and $\mathcal{A}_2$ is given $X_L(S)$ and its goal is to output some set of size $2q + q_p$ such that S is this set. such that S is in this set. We build $\mathcal{A}_2$ as follows:

1. $\mathcal{A}_2$ chooses $i \xleftarrow{\$} \{1, \dots, q_e + q_d\}$.
2. If $\mathcal{A}$ makes a challenge query, $\mathcal{A}_2$ merely simulates Π or Π^{-1} , maintaining consistency with G4.
3. If $\mathcal{A}$ makes its j -th non-challenge query:
 - If $j \stackrel{?}{=} i$ then $\mathcal{A}$ changes the output of X to $X(S)$ when simulator Π (or Π^{-1}).
 - If $j \neq i$, then $\mathcal{A}$ just simulates Π_L or Π_L^{-1} faithfully according to G4.

$\mathcal{A}_2$ outputs the set of all the values of M_l , C_l and the inputs corresponding to θ in offline queries to X_P as its set of guesses. Unless S appears as the input to X is any faithfully simulated query, the two games (G4 and G5) are identical. Thus, the probability that B5 is triggered in G4 or G5 is identical unless such collision occurs.

For the sake of the proof, we introduce the following notation:

- **SB5**: the event that B5 is triggered in the simulated game.
- coll_θ : the event that a seed sampled by the challenger of **GuessL1** collides with a simulated seed.

We have the following,

$$\begin{aligned}
 |\Pr[E_4] - \Pr[E_5]| &\leq \sum_{i=1}^{q_e+q_d} \Pr[\text{B5 at the } i^{\text{th}} \text{ query in G4}] \\
 &\leq (q_e + q_d) \Pr[\text{coll}_\theta] + \sum_{i=1}^{q_e+q_d} \Pr[\text{SB5 at the } i^{\text{th}} \text{ query in G4}] \leq \\
 &\leq \frac{(q_e + q_d)}{2^n} + (q_e + q_d) \text{Adv}^{\text{GuessL1}[2q+q_p]}(\mathcal{A}_2).
 \end{aligned}$$

Essentially, we assume that the first non-challenge query for which $\mathcal{A}$ triggers such bad event is i and $\mathcal{A}_2$ attempts to guess i .

G6: We define the bad event B6: Let θ_i be the input to X in the i^{th} challenge query. $\exists 1 < j < q_p$, s.t. $\theta_j = \theta_i$, where the adversary has made the offline query (θ_i, N) to X_P prior to the i^{th} forgery attempt. From the description of the game up to this point, θ_i is sampled uniformly at random and never appeared before. Thus,

$$|\Pr[E_5] - \Pr[E_6]| \leq \sum_{1 \leq i \leq q_p} \Pr[\text{B6}] \leq \frac{(q_{ce} + q_{cd})q_p}{2^n}.$$

G7: We replace the outputs of X in the challenge queries with random strings. This is equivalent $\$$. Thus, there exists a **PRG ℓ 1** adversary $\mathcal{A}_3$ such that

$$|\Pr[E_6] - \Pr[E_7]| \leq \text{Adv}_X^{\text{PRG}\ell 1}(\mathcal{A}_3),$$

where $\mathcal{A}_3$ makes $q_{ce} + q_{cd}$ online challenge queries, $q_e + q_d$ non-challenge queries and q_p offline queries to X_P .

G8: We note that all the challenge queries are uniformly distributed in G7, and we can replace them with random strings.

$$|\Pr[E_7] - \Pr[E_8]| = 0.$$

G9: We replace the random strings in challenge queries with strings samples according to the ideal world TURP $\tilde{R}$. We apply [40, Lemma 1]

$$|\Pr[E_8] - \Pr[E_9]| \leq \frac{q(q-1)}{2^{l_{\min}} + 1} \leq \frac{q^2}{2^n},$$

where $l_{\min} > n$ is the minimum plaintext length in challenge queries.

G10, G11 and G12: Now the challenge queries are sampled according to the ideal world. Nevertheless, when we changed $\tilde{E}_K$, first to a TURP in G1 and then to a random function in G2, we modified the leakage oracles. Therefore, G9 is not identical to the ideal world. We only need to undo these changes. Arguing exactly the same is in those games we get:

$$\begin{aligned} |\Pr[E_9] - \Pr[E_{12}]| &\leq \text{Adv}_{\tilde{E}}^{\text{stprp}}(\mathcal{A}_5) + \frac{(2(q_e + q_d))^2}{2^n} \\ &\leq \text{Adv}_{\tilde{E}}^{\text{stprp}}(\mathcal{A}_1) + \frac{(2(q_e + q_d))^2}{2^n}, \end{aligned}$$

where the last inequality holds because $\mathcal{A}_5$ makes less queries than $\mathcal{A}_1$. Finally, G12 is identically to the (collision-free) ideal world, which concludes the proof. $\square$

$\square$

Clarifying the proof model: In the proof of Theorem 5, the adversary has access to the ideal primitive X . As such, this is classified as a Random-Oracle-Model (ROM) proof. However, in the proof it is emphasized that the two standard-model primitives $\tilde{E}$ and H (especially $\tilde{E}$) do not have access to X , and are sampled independently. If H is not independent of X , then this will affect the GuessL1 and PRGL1 reductions, but this can be taken care of by a dedicated analysis. Similarly, If $\tilde{E}$ is not independent of X , a dedicated hybrid analysis is needed to ensure that the dependency does not lead to a security break.

The GuessL1, PRGL1 and PRGL1 security of X cannot be quantified except in the black-box model. In the leakage-resilient model, the security relies on both the cryptanalysis of X and scrutinizing it using SPA-style side-channel analysis. We would like to clarify that the reduction can be made simpler by assuming that H is also a random oracle, independent of X , rather than just a collision-resistant hash function. We avoid that for two reasons. A random oracle assumption is a much stronger assumption and in fact several standardized hash functions are not indifferentiable from a random oracle, such as SHA2. Besides, X is also not exactly a random oracle but something slightly weaker, since the online queries only include random inputs. Adding a random oracle assumption on H is stronger than that, as the adversary can control and choose the inputs during online queries.

5.1 The necessity of hard-to-invert leakage

It maybe tempting to consider the possibility of defining X with unbounded leakage. However, leaking the intermediate states of a wide-block cipher has lead to negative results in the literature. For instance, in the case of Feistel ciphers, it was shown that leaking the intermediate states mandate a super-logarithmic number of bounds in order to proof security [27]. In our context, leaking The input to X is a *disastrous* idea. Assume that $\mathcal{A}$ is an adversary that makes a query (M_l, M_r) and in return gets the ciphertext (C_l, C_r) and the input to X , which be denote by θ . For simplicity, we fix (N, A) and omit them in this analysis. Next, the adversary makes an offline query to X with C_l as input and an online challenge query to the construction with (θ, C_r) as the plaintext. The adversary can clearly tell whether the challenge query has been generated by X or by an independent random permutation. The attack does not work if we use different TBCs for different rounds of the construction, making it harder for the adversary to force the input of X to a predictable value. It is worth exploring whether using different TBCs can lead to relaxed assumptions on the leakage of X for STPRP ℓ 2 security. However, since we need this assumption later for other security notions anyway, we leave this exploration outside the scope of this paper. Besides, even if the attack is inapplicable, the PRG ℓ 1 assumption takes into consideration the uniformity and secrecy of the inputs.

5.2 The infeasibility of challenge leakage with unbounded hash leakage

As discussed in [8, 10], it is considered impossible to achieve confidentiality with challenge leakage in the deterministic setting (nonce-misuse in the case of AEAD) without having strong countermeasures. Consider for instance an adversary that makes the challenge query $(N, A, (M_l, M_r))$. The adversary can observe the leakage for that query and try to match it with the leakage from a previous query $(N, A, (M_l, M_r^*))$ where M_r and M_r^* differ in only a few bits. Thus, achieving a notion stronger than STPRP ℓ 2 (in general) is considered impossible. This also applies for the CCAML2 AEAD notion. However, we observe that we it can be possible to prove a result slightly stronger than STPRP ℓ 2, but weaker than STPRP with challenge leakage: instead of PRG ℓ 1, we can rely on PRGL1 and release the leakage from X during challenge queries, while $\tilde{E}$ and the hash function are leak-free. This is unlikely to give more efficient constructions, so we leave it out of scope. On the other hand, as discussed in [8], achieving CCAmL2 for AEAD is feasible, where the challenge leakage uses only fresh nonces. One would hope that STPRP ℓ 2 can be used to achieve it, and consequently the *Motte-and-Bailey* framework would achieve the best feasible notions. Unfortunately, this is not the case, at least not generically. While X can be leakage-resilient for fresh inputs, the inclusion of the hash function H and how it processes plaintexts adds leakages can break confidentiality. Thus, such notion can be reached if and only if H achieves what is known as Left-Or-Right-Leakage resilience (LORL2) [10]. Informally, this notion means that an adversary cannot distinguish between the

leakage resulting from processing M_0 and M_1 , where M_0 and M_1 are chosen by the adversary. If H has no secret material, the only hope to achieve this notion is to apply strong countermeasures. Thus, the **Motte-and-Bailey** framework cannot reach this notion generically. We shall give a candidate that achieves this notion by including a secret function in H that is used once and does not affect its collision resistance, but provides enough randomness to make sophisticated side-channel attacks unfeasible.

5.3 Improved CIML2 security

While Theorem 3 shows that STPRP ℓ 2 implies CIML2, the reduction is quite loose. For instance, the reduction implies the terms

$$q_f \epsilon_{CR} \text{ and } q_f \text{Adv}_{\tilde{E}}^{\text{STPRP}}(\mathcal{A}).$$

These terms are uncharacteristic and indeed not tight. In this section, we give a more dedicated security proof for CIML2. In the remainder of the paper, we consider the following encoding scheme that satisfies our requirements: the plaintext M is at least n . We split M as

$$(M_l, M_r) \xleftarrow{n,*} M$$

where M_l is the first (leftmost) n bits of the plaintext, and M_r is the remainder of the plaintext. The plaintext input to MaB is given as

$$(M_l, 0^t \| M_r).$$

Theorem 6. *Let H be an (ϵ_{CR}, t') -collision-resistant hash function with unbounded leakage, $\tilde{E}$ is a leak-free TBC, and X is a leakage-resilient random function, where $\tilde{E}$ and H are sampled independently of X . Then, for any adversary $\mathcal{A}$ against EtE [MaB, t] that makes q_e encryption queries, q_f forgery attempts, and q_p offline queries to X , such that $\mathcal{A}$ runs in time at most t and $t' > t + (2q_e + q_f)t_H$:*

$$\begin{aligned} \text{Adv}_{\text{EtE}[\text{MaB}, t]}^{\text{CIML2}}(\mathcal{A}) &\leq \text{Adv}_{\tilde{E}}^{\text{STPRP}}(\mathcal{A}_1) + \epsilon_{CR} \\ &\quad + \frac{3(2q_e + q_f)^2 + q_f q_p}{2^n} + \frac{q_f}{2^t} \\ &\quad + 2q_e \text{Adv}_X^{\text{GuessL1}[q_f]}(\mathcal{A}_3) + \text{Adv}_X^{\text{PRGL1}}(\mathcal{A}_4). \end{aligned}$$

where $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$ and $\mathcal{A}_4$ are any adversaries against their respective games. $t' > t + (2q_e + q_f)t_H$, where t_H is the time of one call to H .

The proof of Theorem 6 follows similar structure to that of Theorem 5. The goal is to show that as long as θ is not leaked in any query, the values that are compared to 0^t are indistinguishable from random under the PRGL1 assumption. The complete proof is deferred to Appendix B.3.

Tightness of the CIML2. In the proof of CIML2, we are being generous by considering any repetition of θ to be a win for the adversary, whether this repetition comes from two online queries, or an online query and an offline query to X . While this approach is tight against STPRP ℓ 2, it is not clear that it is always tight against CIML2. For instance, the adversary can guess θ using offline queries, but it still needs to find the corresponding ciphertext to produce this θ , and that ciphertext needs to satisfy that the first t bits are the same as the first t bits when X is queried with that θ and the same nonce. One possible attack under the assumption that X has unbounded leakage is to guess a θ such that the first t bits of $X(\theta, N)$ are 0^t . Then, query a message with N as the nonce and $M_l = \theta$. We can then leak the input of X and we get a valid forgery. This attack is non-trivial, since the first step requires 2^t offline computations and a weaker assumption on X . However, t can be a lot smaller than n , and if $t < n/2$ then the attack is better than generic attacks. Thus, we do not expect a significant improvement with a finer analysis than what we provide.

On the other hand, we note that Theorem 6 is indeed an improvement over Theorem 3. To see that, consider $t \leq n/2$, which is quite a reasonable assumption if $n = 256$ and $t = 128$. Consider also that the hash output is 256 bits. Theorem 3 provides strictly less than 86 bits of security, due to the existence of cubic terms. Theorem 6 ensures 128-bit security, with only quadratic terms with respect to n and the hash output, but only a linear term t .

Last but not least, we note that the term $\text{Adv}_X^{\text{PRGL1}}(\mathcal{A}_4)$ may also not be tight. The adversary cannot use the leakage from challenge queries (in this case forgeries) except for future queries. Thus, PRG ℓ 1 maybe a sufficient assumption. However, a proof with PRG ℓ 1 would give a bound $q_f \text{Adv}_X^{\text{PRG}\ell 1}(\mathcal{A}_4)$ using a hybrid argument. Schemes such as the Sponge construction can target both assumptions, and it is likely that the bound we get from PRGL1 is better since it does not rely on this hybrid argument.

6 Security Requirements for Challenge Leakage

In this section, we present a strengthened variant that achieves all the same security goals, as well as CCAmL2. Thus, it achieves the best possible AEAD security (up to the birthday bound). The strengthened variant is described in Algorithm 1 (right), where the hash function includes a preprocessing stage that permutes the nonce N using a secret leak-free permutation. This permutation provides enough randomness in the hash function, while making sure that the public part of the hash function H (which is leaky) always absorbs a fresh, unique, secret first block γ . This allows proving CCAmL2. In order to do so, we need to define the challenge leakage, which did not exist so far. The leakages from X and $\tilde{E}_K$ are the same as before. For H and $\oplus$, we need to define the Left-or-Right security with challenge Leakage (LORL2), similar to the notion used in [10].

Definition 1. Let $f : \mathcal{S} \times \mathcal{M} \rightarrow \mathcal{C}$ be a function. We define the advantage of and adversary $\mathcal{A}$ against LORL2 as:

$$\begin{aligned} \text{Adv}_f^{\text{LORL2}}(\mathcal{A}) &\stackrel{\text{def}}{=} \\ &|\Pr[s \xleftarrow{\$} \mathcal{S} : \mathcal{A}(f(s, m_0), L(f(s, m_0))) \Rightarrow 1] - \\ &\Pr[s \xleftarrow{\$} \mathcal{S} : \mathcal{A}(f(s, m_1), L(f(s, m_1))) \Rightarrow 1]| \end{aligned}$$

As discussed in [10], it is hard to achieve such notion without some physical countermeasures. However, the purpose of this notion is rather establishing pre-scriptive: it shows that the implementer needs to dedicate some resources to study the implementation of the function f . Besides, countermeasures for SPA attacks that can be used to break this notion are usually cheaper than DPA countermeasures.

6.1 Combined LORL2 security

We described the individual games for ease of understanding, but in our analysis we require a combined notion

1. The challenger selects a random bit $b \xleftarrow{\$} \{0, 1\}$.
2. The adversary $\mathcal{A}$ sends (N, A, Y_{r_0}, Y_{r_1}) .
3. The challenger chooses a random seed θ , and returns $X(\theta, N) \oplus Y_{r_b}$.
4. $\mathcal{A}$ receives the leakage from the challenger's output, $L(H(N \| A \| A \| X(\theta, N) \oplus Y_{r_b}))$, $L(H(N \| A \| A \| X(\theta, N) \oplus Y_{r_b}))$ and $L(X(\theta, N) \oplus Y_{r_b})$.
5. $\mathcal{A}$ returns a bit b' .

We shall define this game simply as LORL2, and we define $\mathcal{A}$'s advantage as

$$\text{Adv}_{\text{comb}}^{\text{LORL2}}(\mathcal{A}) \stackrel{\text{def}}{=} |\Pr[\mathcal{A}^{\text{LORL2}} \Rightarrow 1 | b = 1] - \Pr[\mathcal{A}^{\text{LORL2}} \Rightarrow 1 | b = 0]|.$$

Theorem 7. Let H be an (ϵ_{CR}, t') -collision-resistant hash function with unbounded leakage, $\tilde{E}$ is a leak-free TBC, and X is a leakage-resilient random function, where $\tilde{E}$ and H are sampled independently of X . Then, for any adversary $\mathcal{A}$ against EtE [MaB, t] that makes q_e encryption queries, q_f forgery attempts, q_c challenge queries, and q_p offline queries to X , such that $\mathcal{A}$ runs in time at most t :

$$\begin{aligned} \text{Adv}_{\text{EtE}[\text{MaB}, t]}^{\text{CCAML2}}(\mathcal{A}) &\leq \text{Adv}_{\text{EtE}[\text{MaB}, t]}^{\text{CIML2}}(\mathcal{A}_1) + \text{Adv}_{\tilde{E}}^{\text{STPRP}}(\mathcal{A}_2) + q_c \text{Adv}_X^{\text{GuessL1}[q_p]}(\mathcal{A}_3) \\ &\quad + \text{Adv}_X^{\text{PRGL1}}(\mathcal{A}_4) + q_c \text{Adv}_{\text{comb}}^{\text{LORL2}}(\mathcal{A}_5) + \epsilon_{CR} + \frac{2(2q_e + q_f)^2}{2^n}, \end{aligned}$$

where $t' > t + 2qt_H$ and t_H is the time of one call to H .

The proof of Theorem 7 is deferred to Appendix B.4. The proof requires that H is secure against two-trace attacks in practice, as for any fresh nonce, the adversary gets two traces for the hash function evaluated with the this nonce. Under this assumption, the strengthened MaB achieves the strongest practically feasible combination of notions:

$$\text{STPRP}\ell 2 \wedge \text{CIML2} \wedge \text{CCAmL2} \wedge \text{CCAML2}.$$

7 Qaitbay: A Sponge-Based Instantiation

In this section we instantiate the **Motte-and-Bailey** framework using two independent primitives: the **Sponge** construction and an independently selected TBC. This instantiation is described in Construction 1 and Algorithm 2.

Construction 1 Let n, k, b, h, r, c be positive integers such that $b = r + c$ and $h \geq c$. Let $\tilde{E} : \{0, 1\}^k \times (\{0, 1\}^h \setminus \mathcal{S}_b) \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be an *STPRP*-secure TBC, and $P : \{0, 1\}^b \rightarrow \{0, 1\}^b$ be an independent publicly-accessible random permutation. Then, the **Qaitbay** family of accordion modes is given by Algorithm 2. **Sponge**: $\{0, 1\}^* \times \mathbb{Z}^+ \rightarrow \{0, 1\}^*$ is the sponge construction defined in [14] with rate r , capacity c and permutation size b . Its second input defines the length of its output. $\mathcal{S}_b$ is a set of forbidden tags that $\tilde{E}$ is not allowed to use.

In the remainder of this section, we study the concrete security of **Qaitbay** and its performance, giving plausible practical instantiations.

Corollary 1. Let the **Qaitbay-2** construction be defined as described in Construction 1. Let σ be the sum of the number of r -bit blocks processed by **Sponge**(0||·) in all queries. Let σ' be the number of r -bit blocks in M_r in all queries. Let q_p be the number of primitive calls to the permutation used in **Sponge**. Then, for any adversary $\mathcal{A}$ that runs in time at most t and makes q_e, q_d, q_{ce}, q_{cd} queries, as defined in Theorem 5, there exists an adversary $\mathcal{A}_1$ that makes $2(q_e + q_d)$ queries to $\tilde{E}$, and four adversaries $\mathcal{A}_2, \mathcal{A}_3, \mathcal{A}_4$, and $\mathcal{A}_5$, such that

$$\begin{aligned} \text{Adv}_{\text{Qaitbay-2}[\tilde{E}, \text{Sponge}]}^{\text{STPRP} \ell 2}(\mathcal{A}) &\leq 2\text{Adv}_{\tilde{E}}^{\text{STPRP}}(\mathcal{A}_1) + \sigma \text{Adv}_{\text{P}}^{\text{GuessL1}[q_g]}(\mathcal{A}_2) \\ &\quad + \text{Adv}_{\text{sponge}}^{\text{coll}}(\mathcal{A}_3) + (|\mathcal{S}_b| + 1) \text{Adv}_{\text{sponge}}^{\text{pre}}(\mathcal{A}_4) \\ &\quad + \text{Adv}_{\text{sponge}}^{\text{PRG}}(\mathcal{A}_5) + \frac{22q^2}{2^n}, \end{aligned}$$

Algorithm 2 Qaitbay-1 (left) and Qaitbay-2 (right). Lines 4, 9, on the left-hand side and lines 1, 5 and 10 on the right-hand side, are leak-free. For further clarity, this lines have been coloured teal.

1: $L \leftarrow 0 \ N \ \ A \ \ A \ M_r$	1: $\gamma \leftarrow \tilde{E}_K^1(N)$
2: $L \leftarrow L \ 1 \ 0^{r-1-(L \bmod r)}$	2: $L \leftarrow 0 \ \gamma \ \ A \ \ A \ M_r$
3: $\alpha \leftarrow \text{Sponge}^{b,r,c}(L, h)$	3: $L \leftarrow L \ 1 \ 0^{r-1-(L \bmod r)}$
4: $\theta \leftarrow \tilde{E}_K^\alpha(M_l)$	4: $\alpha \leftarrow \text{Sponge}^{b,r,c}(L, h)$
5: $C_r \leftarrow M_r \oplus \text{Sponge}^{b,r,c}(1 \ N \ \theta, M_r)$	5: $\theta \leftarrow \tilde{E}_K^\alpha(M_l)$
6: $L \leftarrow \text{ftarrow} 0 \ N \ \ A \ \ A \ C_r$	6: $C_r \leftarrow M_r \oplus \text{Sponge}^{b,r,c}(1 \ N \ \theta, M_r)$
7: $L \leftarrow L \ 1 \ 0^{r-1-(L \bmod r)}$	7: $L \leftarrow 0 \ \gamma \ \ A \ \ A \ C_r$
8: $\alpha \leftarrow \text{Sponge}^{b,r,c}(L, h)$	8: $L \leftarrow L \ 1 \ 0^{r-1-(L \bmod r)}$
9: $C_l \leftarrow \tilde{E}_K^\alpha(\theta)$	9: $\alpha \leftarrow \text{Sponge}^{b,r,c}(L, h)$
10: return $C_l \ C_r$	10: $C_l \leftarrow \tilde{E}_K^\alpha(\theta)$
	11: return $C_l \ C_r$

where $\mathcal{A}_3$ and $\mathcal{A}_4$ are collision and preimage finding adversaries, respectively, for $\text{Sponge}(0||\cdot)$, they run in time $O(t)$ and make $2(q_e + q_d)$ queries to Sponge . $\mathcal{A}_5$ is a black-box PRG adversary against the full-state keyed sponge PRF that runs in time $O(t)$ and makes $2\sigma + 2 + q_p$ primitives queries to P . Then, $q_g = q_p + 2\sigma + 2(q_e + q_d)$. $q = q_e + q_d + q_{ce} + q_{cd}$. Let $2\sigma + \sigma' + 3q + q_p < \min\{2^{n/2}, 2^{c/2}\}$.

Proof. Corollary 1 follows from Theorem 5 as follows: First, we build an adversary that simulates the game given oracle access to P and $\tilde{E}_K$. This adversary terminates the game if any hash output is either 1 or in $\mathcal{S}_b$. This is a preimage-finding adversary against the hash function. Next, we replace $\tilde{E}$ with a TURP. Next, we terminate the game if a collision is found at the output of the hash function. Next, we consider an adversary that simulates the construction except X (the middle application of Sponge), and wins if it guesses any of the secret inputs to P . This follows from Guo et al. [38]. Finally, if all the secret inputs to P never appears in the offline queries to P or during the hash function, then any adversary that breaks the PRGL1 security of X also breaks its black-box security. The advantage is given by [58, Theorem 5]. $\square$

Corollary 1 gives us an indication of what security to expect. As long as $|\mathcal{S}_b|$ is small, P is protected against single-trace attacks, and $\tilde{E}$ is leak-free, then Qaitbay is secure as long as (roughly) $(3D + T) < \min\{2^{n/2}, 2^{c/2}\}$. Setting $n = c = 256$ and $|\mathcal{S}_b| \in \{0, 1, 2\}$, we get 128-bit security. We note that the proof for Qaitbay-1 STPRP ℓ 2 security follows exactly the same, with $|\mathcal{S}_b|$ instead of $|\mathcal{S}_b| + 1$, as the constant 1 just captures the condition that the output of the hash function collides with the tweak of the first TBC call. We note that the LORL2 security of Sponge also follows from the analysis of Guo et al. [38]. Thus, Qaitbay-1 and Qaitbay-2 have the same level of CIML2 and CCAM ℓ 2 security (up to 2^t forgeries), and Qaitbay-2 has the same level of CCAM ℓ 2 security.

7.1 Practical Instantiations

Now, we define the concrete instantiations. We give three pairs of instantiations. All instantiations use TurboSHAKE-256 [13] (i.e. with 256-bit capacity). They differ in the instantiation of $\tilde{E}$. The algorithmic description of the three different instantiations of $\tilde{E}$ is given below:

Qaitbay [PHOTON, TurboSHAKE]: The TBC is instantiated using the XPX TBC [57], using the PHOTON-256 permutation. The instantiation requires Galois field multiplication over $\text{GF}(2^{256})$ and is defined by $K' \leftarrow K \odot T$; $C \leftarrow \text{PHOTON}(K' \oplus P) \oplus K'$. Mennink [57] defines a set of valid tweak for which the construction is secure. In our case, the relevant condition is $T \notin \{0, 1\}$, where the latter coincides with the condition also imposed generically by Qaitbay-2. This extends the $\mathcal{S}_b$ to $\{0, 1\}$. The STPRP security of XPX is up to $D \cdot T < 2^n$.

Qaitbay [Rijndael-IC, TurboSHAKE]: We can use an ideal cipher with 256-bit key and 256-bit block to instantiate a TBC with 256-bit key, 256-bit tweak, and

256-bit block, while providing 128-bit security with a straightforward construction

$$C \leftarrow \text{IC}(K \oplus T, M).$$

In practice, a candidate for replacing this ideal cipher is `Rijndael-256-256`.

Qaitbay [Rijndael-PRP, TurboSHAKE]: We can construct a TBC 128-bit key, 256-bit tweak, and 256-bit block using a block cipher with 128-bit key and 256-bit block, using the seminal LRW2 construction [51].

$$L \leftarrow E_{K \oplus 1}(T); C \leftarrow E_K(M \oplus L) \oplus L.$$

This can be instantiated using `Rijndael-256-128`. In terms of security, this instantiation provides the same bit security as the rest (128), but it is more immune against key recovery attacks, and requires less key material. On the flip side, it requires double the number of queries. In case of `Qaitbay-1`, we need 4 calls to the block ciphers. In case of `Qaitbay-2`, we need 5 calls, since processing the tweak for initializing the hash function can be amortized amongst all construction calls. On one hand, this shows that improving security against key recovery comes at a cost. On the other, it demonstrated that `Qaitbay-2` is not significantly more expensive than `Qaitbay-1`.

Rationale: The `PHOTON` family of hash functions was introduced by Guo *et al.* [39] and has been standardized under ISO/IEC 29192-5:2016. It is based on the Sponge function and has been widely analyzed. The underlying permutation, particularly `PHOTON-256` which we use, was also used as part of `PHOTON-Beetle` [3], a finalist in the NIST lightweight cryptography project. Thus, we believe it is a trusted cryptographic permutation. On the other hand, it does not suffer from some of the implementation shortcomings of other cryptography permutations such as `Keccak-p`, where the inverse of the permutation is not efficient to implement. While it is not particularly masking-friendly, its secure implementations have been studied as part of the NIST lightweight project [59].

`Rijndael` [22] is the precursor to the AES block cipher, and in the initial family of block ciphers, one of the variants uses 256-bit key and block. Despite not being standardized, it is based on the same internal primitives and a similar internal structure. The block size suits the use case of `Qaitbay` perfectly, and the fact that it is based on the AES primitives means that its masked implementation inherits a lot of the analysis already performed on AES.

Despite their nice features, instantiations based on `PHOTON` and `Rijndael` suffer from two limitations. Both these primitives were not designed to be masking-friendly, and incur a large cost for protecting against side-channel attacks. Besides, all these instantiations provide 128-bit security but require a 256-bit key. We believe this is acceptable given that these are the first practical leakage-resilient Accordion modes. However, in the long term, we believe a `Qaitbay` variant with a dedicated TBC is an interesting design choice to explore. That TBC would need to be masking-friendly and with parameters: 128-bit key, 256-bit tweak, and 256-bit block.

8 Alicante: A TBC-Based Instantiation

The **Alicante** family is described in Algorithm 3. **Alicante** is designed in the ideal cipher model. First, we need to introduce an assumption on the ideal cipher used in the construction in Algorithm 3.

Definition 2. Let $\tilde{E} : \{0, 1, 2, 3\} \times \{0, 1\}^{t_w} \times \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be an n -bit ideal cipher with the following leakage functions:

1. If $B \in \{0, 3\}$, $L(\tilde{E}_X^{B,Y}(M)) = \perp$.
2. If $B \in \{1\}$, $L(\tilde{E}_X^{B,Y}(M)) = (B, Y, X, M)$.
3. If $B \in \{2\}$, $L(\tilde{E}_X^{B,Y}(M)) = L_{\text{NI}}(B, Y, X, M)$.

Let $c > 1$ be an integer and let $\pi_0 : \{0, 1\}^{cn} \times \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ and $\pi_3 : \{0, 1\}^{cn} \times \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ and be independent leak-free tweakable permutations. Let C be a construction that uses $B \in \{0, 3\}$ with unique key K sampled uniformly at random and that it also uses $B \in \{1, 2\}$ with varying keys. Then, we assume that for any adversary $\mathcal{A}$

$$\begin{aligned} & |\Pr[K \xleftarrow{\$} \{0, 1\}^k : \mathcal{A}^{C[\tilde{E}_K^0, \tilde{E}^1, \tilde{E}^2, \tilde{E}_K^3], \tilde{E}^0, \tilde{E}^1, \tilde{E}^2, \tilde{E}^3} \Rightarrow 1] - \\ & \Pr[\pi_0 \xleftarrow{\$} \text{tperm}(\{0, 1\}^{2n}, \{0, 1\}^n) \pi_3 \xleftarrow{\$} \text{tperm}(\{0, 1\}^{2n}, \{0, 1\}^n) : \\ & \mathcal{A}^{C[\pi_0, \tilde{E}^1, \tilde{E}^2, \pi_3], \tilde{E}^0, \tilde{E}^1, \tilde{E}^2, \tilde{E}^3} \Rightarrow 1]| \leq \frac{q_p}{2^k}, \end{aligned}$$

where q_p is the number of queries the adversary makes to all its oracles, where the adversary also gets the leakage of their queries.

Corollary 2. Let the **Alicante-2** construction be defined as described in Algorithms 3 and 4. Let σ_H be the total number of $(t_w - n)$ -bit blocks processed by H in all queries. Let σ_X be the number of n -bit blocks in M_r in all queries. Let q_p be the number of primitive calls to the ideal cipher. Then, for any adversary

Algorithm 3 Alicante-1 (left) and Alicante-2 (right). Protected TBC calls are colored in teal.

1: $L \leftarrow N A A M_r$	1: $\gamma \leftarrow \tilde{E}_K^{3,0}(N)$
2: $\delta \leftarrow t_w - n - 1 - (L \bmod (t_w - n))$	2: $L \leftarrow \gamma A A M_r$
3: $\alpha \leftarrow \text{MDPH}[\tilde{E}^1](L 1 0^\delta)$	3: $\delta \leftarrow t_w - n - 1 - (L \bmod (t_w - n))$
4: $\theta \leftarrow \tilde{E}_K^{0,\alpha}(M_l)$	4: $\alpha \leftarrow \text{MDPH}[\tilde{E}^1](L 1 0^\alpha)$
5: $C_r \leftarrow M_r \oplus 2\text{PRG}[\tilde{E}^{2,N}](\theta)$	5: $\theta \leftarrow \tilde{E}_K^{0,\alpha}(M_l)$
6: $L \leftarrow N A A C_r$	6: $C_r \leftarrow M_r \oplus 2\text{PRG}[\tilde{E}^{2,N}](\theta)$
7: $\delta \leftarrow t_w - n - 1 - (L \bmod (t_w - n))$	7: $L \leftarrow \gamma A A C_r$
8: $\alpha \leftarrow \text{MDPH}[\tilde{E}^1](L 1 0^\delta)$	8: $\delta \leftarrow t_w - n - 1 - (L \bmod (t_w - n))$
9: $C_l \leftarrow \tilde{E}_K^{0,\alpha}(\theta)$	9: $\alpha \leftarrow \text{MDPH}[\tilde{E}^1](L 1 0^\alpha)$
	10: $C_l \leftarrow \tilde{E}_K^{0,\alpha}(\theta)$

$\mathcal{A}$ that runs in time at most t and makes q_e, q_d, q_{ce}, q_{cd} queries, as defined in Theorem 5,

$$\begin{aligned} \text{Adv}_{\text{Alicante-2}[\tilde{E}]}^{\text{STPRP}\ell 2}(\mathcal{A}) &\leq \frac{q_p}{2^k} + \frac{22q^2 + qq_p}{2^n} \\ &\quad + \frac{6(6\sigma_H + (3+n)q_p)}{(2^n - 4\sigma_H - 2q_p)^2} + \frac{24n\sigma_H + 12nq_p}{2^n - 4\sigma_H - 2q_p} + \frac{(nq_p + 2q)^2}{2^{2n}} \\ &\quad + (q_e + q_d)\text{Adv}_{\tilde{E}}^{2\text{-up}[q_g]}(\mathcal{A}_1) + 2\sigma_X\text{Adv}_{\tilde{E}}^{2\text{-up}[q_g]}(\mathcal{A}_2), \end{aligned}$$

where $q_g = q_{IC} + 4\sigma_h$ and $q = q_e + q_d + q_{ce} + q_{cd}$.

The proof follows from Theorem 5 and the techniques of Berti *et al.* [10]. It is deferred to Appendix B.5.

8.1 Concrete Instantiation: Deoxys-128-640

Deoxys-128-640 is a *large-tweak* TBC proposed by Cogliati *et al.* [19] as a variant of the Deoxys-TBC [43]. The details of the design are not relevant to our paper, but the design is based on the AES round function and consists of 20 such rounds, alongside a tweakkey schedule that takes 640 bits as input and generates 21 128-bit round keys. In this section, we provide two instances of Alicante

Algorithm 4 Alicante-2 [Deoxy-128-640]. In the case of Alicante-1, the first TBC calls is removed and all instances of γ are replaced by N . Protected TBC calls are colored in teal.

1: $\gamma \leftarrow \tilde{E}_K(11, 0^{638}, N)$ 2: $L \leftarrow \gamma \ A \ A \ M_r$ 3: $L_1, \dots, L_l \xleftarrow{L} L \ 1 \ 0^{637 - (L \bmod 638)}$ 4: $i \leftarrow 0$ 5: $(H_0, H_1) \leftarrow (0^{128}, 0^{128})$ 6: while $i < l$: 7: $Z \leftarrow \tilde{E}_{H_1}(01, L_i, H_0) \oplus H_0$ 8: $H_1 \leftarrow \tilde{E}_{H_1}(01, L_i, H_0 \oplus 0^{127} \ 1) \oplus H_0 \oplus 0^{127} \ 1$ 9: $H_0 \leftarrow Z$ 10: $H_0 \leftarrow H_0 \oplus 0^{126} \ 10$ 11: $Z \leftarrow \tilde{E}_{H_1}(01, L_l, H_0) \oplus H_0$ 12: $H_1 \leftarrow \tilde{E}_{H_1}(01, L_l, H_0 \oplus 0^{127} \ 1) \oplus H_0 \oplus 0^{127} \ 1$ 13: $H_0 \leftarrow Z$ 14: $\theta \leftarrow \tilde{E}_K(00, 0^{382} \ H_0 \ H_1, M_i)$ 15: $M_1, \dots, M_m \xleftarrow{128} M_r$ 16: $\theta' \leftarrow \theta$ 17: while $i < m$: 18: $C_i \leftarrow M_i \oplus \tilde{E}_{\theta'}(10, 0^{510} \ N, 0^{128})$	19: $\theta' \leftarrow \tilde{E}_{\theta'}(10, 0^{510} \ N, 0^{127} \ 1)$ 20: $C_m \leftarrow M_m \oplus \lfloor \tilde{E}_{\theta'}(10, 0^{510} \ N, 0^{128}) \rfloor_{ M_m }$ 21: $C_r \leftarrow C_1 \ \dots \ C_m$ 22: $L \leftarrow \gamma \ A \ A \ C_r$ 23: $L_1, \dots, L_l \xleftarrow{L} L \ 1 \ 0^{637 - (L \bmod 638)}$ 24: $i \leftarrow 0$ 25: $(H_0, H_1) \leftarrow (0^{128}, 0^{128})$ 26: while $i < l$: 27: $Z \leftarrow \tilde{E}_{H_1}(01, L_i, H_0) \oplus H_0$ 28: $H_1 \leftarrow \tilde{E}_{H_1}(01, L_i, H_0 \oplus 0^{127} \ 1) \oplus H_0 \oplus 0^{127} \ 1$ 29: $H_0 \leftarrow Z$ 30: $H_0 \leftarrow H_0 \oplus 0^{126} \ 10$ 31: $Z \leftarrow \tilde{E}_{H_1}(01, L_l, H_0) \oplus H_0$ 32: $H_1 \leftarrow \tilde{E}_{H_1}(01, L_l, H_0 \oplus 0^{127} \ 1) \oplus H_0 \oplus 0^{127} \ 1$ 33: $H_0 \leftarrow Z$ 34: $C_l \leftarrow \tilde{E}_K(00, 0^{382} \ H_0 \ H_1, \theta)$ 35: return (C_l, C_r)
---	--

with $n = k = 128$, and $t_w = 638$. The security is maintained as long as the total amount of data encrypted blocks D and the total number of primitive queries T satisfy $DT < 2^{128}$, and $D < 2^{64}$. The complete instantiation is given in Algorithm 4. We note that we can optimize the scheme further by choosing different variants for **Deoxys-TBC** in different purposes, instead of always padding the tweak parameters to 640 bits. However, what we present in this subsection is merely a proof of concept on the instantiability **Alicante**, so for the sake of simplicity of the presentation, we use only one variant.

9 Conclusion

In this work we have studied the relation between **STPRP ℓ 2** security and **CIML2/CCAM ℓ 2** security using the generic **EtE** framework. We have introduced **Motte-and-Bailey**, a novel, generic framework to build secure leakage-resilient accordion modes. Furthermore, we have shown said framework with four competitive instantiations, with strong security properties and bounds, for **STPRP ℓ 2**, **CIML2** and **CCAM ℓ 2**, and **CCAM12**. Finally, we have argued that these instantiations, **Qaitbay** and **Alicante** are more efficient for the same level of security to previous works in the leakage-resilience setting.

While **Motte-and-Bailey** is a promising framework, there are still future research directions to be explored. One problem is building **Qaitbay** using a single permutation. Not only would the proof have to be changed, but also this permutation has to be easily invertible to be used in **XPX**. We also note that **XPX** may not be the only approach to instantiate $\tilde{E}$ from a public permutation. The second problem is studying the security of **MaB** when $\tilde{E}$ is not leak-free, but only **STPRP ℓ 2** secure. This problem in particular will be a step further in this domain. Last but not least, both **Qaitbay** and **Alicante** can benefit from a masking-friendly TBC with 128-bit key, 256-bit block and more than 256-bit tweak.

Acknowledgement

This work is partially funded by the Wallenberg-NTU Presidential Postdoctoral Fellowship. The project was performed while Mario Marhuenda Beltrán was on a research attachment at NTU, Singapore. This work is also funded by the Singapore MoE Tier 1 grant RG111/25.

References

1. Andreeva, E., Bhati, A.S., Preneel, B., Vizár, D.: 1, 2, 3, Fork: Counter Mode Variants based on a Generalized Forkcipher. *IACR Trans. Symmetric Cryptol.* **2021**(3), 1–35 (2021). <https://doi.org/10.46586/TOSC.V2021.I3.1-35>, <https://doi.org/10.46586/tosc.v2021.i3.1-35>

2. Bacuieti, N., Daemen, J., Hoffert, S., Assche, G.V., Keer, R.V.: Jammin' on the Deck. In: Agrawal, S., Lin, D. (eds.) *Advances in Cryptology - ASIACRYPT 2022 - 28th International Conference on the Theory and Application of Cryptology and Information Security*, Taipei, Taiwan, December 5-9, 2022, Proceedings, Part II. *Lecture Notes in Computer Science*, vol. 13792, pp. 555–584. Springer (2022). https://doi.org/10.1007/978-3-031-22966-4_19, https://doi.org/10.1007/978-3-031-22966-4_19
3. Bao, Z., Chakraborti, A., Datta, N., Guo, J., Nandi, M., Peyrin, T., Yasuda, K.: PHOTON-beetle authenticated encryption and hash family. *NIST Lightweight Compet. Round 1*, 115 (2019), <https://csrc.nist.gov/csrc/media/Projects/Lightweight-Cryptography/documents/round-1/spec-doc/PHOTON-Beetle-spec.pdf>
4. Barwell, G., Martin, D.P., Oswald, E., Stam, M.: Authenticated Encryption in the Face of Protocol and Side Channel Leakage. In: Takagi, T., Peyrin, T. (eds.) *Advances in Cryptology - ASIACRYPT 2017 - 23rd International Conference on the Theory and Applications of Cryptology and Information Security*, Hong Kong, China, December 3-7, 2017, Proceedings, Part I. *Lecture Notes in Computer Science*, vol. 10624, pp. 693–723. Springer (2017). https://doi.org/10.1007/978-3-319-70694-8_24, https://doi.org/10.1007/978-3-319-70694-8_24
5. Bellare, M., Namprempre, C.: Authenticated Encryption: Relations among Notions and Analysis of the Generic Composition Paradigm. In: Okamoto, T. (ed.) *Advances in Cryptology - ASIACRYPT 2000, 6th International Conference on the Theory and Application of Cryptology and Information Security*, Kyoto, Japan, December 3-7, 2000, Proceedings. *Lecture Notes in Computer Science*, vol. 1976, pp. 531–545. Springer (2000). https://doi.org/10.1007/3-540-44448-3_41, https://doi.org/10.1007/3-540-44448-3_41
6. Bellare, M., Rogaway, P.: Encode-Then-Encipher Encryption: How to Exploit Nonces or Redundancy in Plaintexts for Efficient Cryptography. In: Okamoto, T. (ed.) *Advances in Cryptology - ASIACRYPT 2000, 6th International Conference on the Theory and Application of Cryptology and Information Security*, Kyoto, Japan, December 3-7, 2000, Proceedings. *Lecture Notes in Computer Science*, vol. 1976, pp. 317–330. Springer (2000). https://doi.org/10.1007/3-540-44448-3_24, https://doi.org/10.1007/3-540-44448-3_24
7. Bellizia, D., Berti, F., Bronchain, O., Cassiers, G., Duval, S., Guo, C., Leander, G., Leurent, G., Levi, I., Momin, C., Pereira, O., Peters, T., Standaert, F., Udvarhelyi, B., Wiemer, F.: Spook: Sponge-Based Leakage-Resistant Authenticated Encryption with a Masked Tweakable Block Cipher. *IACR Trans. Symmetric Cryptol.* **2020**(S1), 295–349 (2020). <https://doi.org/10.13154/TOSC.V2020.IS1.295-349>, <https://doi.org/10.13154/tosc.v2020.is1.295-349>
8. Bellizia, D., Bronchain, O., Cassiers, G., Grosso, V., Guo, C., Momin, C., Pereira, O., Peters, T., Standaert, F.: Mode-Level vs. Implementation-Level Physical Security in Symmetric Cryptography - A Practical Guide Through the Leakage-Resistance Jungle. In: Micciancio, D., Ristenpart, T. (eds.) *Advances in Cryptology - CRYPTO 2020 - 40th Annual International Cryptology Conference*, CRYPTO 2020, Santa Barbara, CA, USA, August 17-21, 2020, Proceedings, Part I. *Lecture Notes in Computer Science*, vol. 12170, pp. 369–400. Springer (2020). https://doi.org/10.1007/978-3-030-56784-2_13, https://doi.org/10.1007/978-3-030-56784-2_13
9. Berti, F., Guo, C., Pereira, O., Peters, T., Standaert, F.: TEDT, a Leakage-Resist AEAD Mode for High Physical Security Applications. *IACR Trans.*

- Cryptogr. Hardw. Embed. Syst. **2020**(1), 256–320 (2020). <https://doi.org/10.13154/TCHES.V2020.I1.256-320>, <https://doi.org/10.13154/tches.v2020.i1.256-320>
10. Berti, F., Guo, C., Pereira, O., Peters, T., Standaert, F.X.: Tedt, a leakage-resist aead mode for high physical security applications. IACR Transactions on Cryptographic Hardware and Embedded Systems **2020**(1), 256–320 (Nov 2019). <https://doi.org/10.13154/tches.v2020.i1.256-320>, <https://tches.iacr.org/index.php/TCHES/article/view/8400>
 11. Berti, F., Guo, C., Peters, T., Standaert, F.: Efficient Leakage-Resilient MACs Without Idealized Assumptions. In: Tibouchi, M., Wang, H. (eds.) Advances in Cryptology - ASIACRYPT 2021 - 27th International Conference on the Theory and Application of Cryptology and Information Security, Singapore, December 6–10, 2021, Proceedings, Part II. Lecture Notes in Computer Science, vol. 13091, pp. 95–123. Springer (2021). https://doi.org/10.1007/978-3-030-92075-3_4, https://doi.org/10.1007/978-3-030-92075-3_4
 12. Bertoni, G., Daemen, J., Hoffert, S., Peeters, M., Assche, G.V., Keer, R.V.: Farfalle: Parallel Permutation-based Cryptography. IACR Trans. Symmetric Cryptol. **2017**(4), 1–38 (2017), <https://tosc.iacr.org/index.php/ToSC/article/view/801>
 13. Bertoni, G., Daemen, J., Hoffert, S., Peeters, M., Assche, G.V., Keer, R.V., Viguier, B.: TurboSHAKE. IACR Cryptol. ePrint Arch. p. 342 (2023), <https://eprint.iacr.org/2023/342>
 14. Bertoni, G., Daemen, J., Peeters, M., Van Assche, G.: Sponge Functions. In: ECRYPT hash workshop. vol. 2007 (2007), <https://csrc.nist.gov/groups/ST/hash/documents/JoanDaemen.pdf>
 15. Bhati, A.S., Verbaauwhede, M., Andreeva, E.: Breaking, Repairing and Enhancing XCBv2 into the Tweakable Enciphering Mode GEM. IACR Cryptol. ePrint Arch. p. 1554 (2024), <https://eprint.iacr.org/2024/1554>
 16. Chakraborty, D., Nandi, M.: An Improved Security Bound for HCTR. In: Nyberg, K. (ed.) Fast Software Encryption, 15th International Workshop, FSE 2008, Lausanne, Switzerland, February 10–13, 2008, Revised Selected Papers. Lecture Notes in Computer Science, vol. 5086, pp. 289–302. Springer (2008). https://doi.org/10.1007/978-3-540-71039-4_18, https://doi.org/10.1007/978-3-540-71039-4_18
 17. Chen, Y.L., Davidson, M., Dworkin, M., Kang, J., Kelsey, J., Sasaki, Y., Turan, M.S., Chang, D., Mouha, N., Largo, M., et al.: Proposal of Requirements for an Accordion Mode. In: Discussion Draft for the NIST Accordion Mode Workshop 2024 (2024), <https://csrc.nist.gov/files/pubs/other/2024/04/10/proposal-of-requirements-for-an-accordion-mode-dis/iprd/docs/proposal-of-requirements-for-an-accordion-mode-discussion-draft.pdf>
 18. Chen, Y.L., Hiraga, Y., Mouha, N., Naito, Y., Sasaki, Y., Sugawara, T.: Beyond-Birthday-Bound Security with HCTR2: Cascaded Construction and Tweak-Based Key Derivation. In: Hanaoka, G., Yang, B. (eds.) Advances in Cryptology - ASIACRYPT 2025 - 31st International Conference on the Theory and Application of Cryptology and Information Security, Melbourne, VIC, Australia, December 8–12, 2025, Proceedings, Part I. Lecture Notes in Computer Science, vol. 16245, pp. 3–34. Springer (2025). https://doi.org/10.1007/978-981-95-5018-0_1, https://doi.org/10.1007/978-981-95-5018-0_1
 19. Cogliati, B., Jean, J., Peyrin, T., Seurin, Y.: A Long Tweak Goes a Long Way: High Multi-user Security Authenticated Encryption from Tweakable Block Ci-

- phers. IACR Communications in Cryptology **1**(2) (2024). <https://doi.org/10.62056/a3qjp2fgx>
20. Crowley, P., Biggers, E.: Adiantum: Length-Preserving Encryption for Entry-Level Processors. IACR Trans. Symmetric Cryptol. **2018**(4), 39–61 (2018). <https://doi.org/10.13154/TOSC.V2018.I4.39-61>, <https://doi.org/10.13154/tosc.v2018.i4.39-61>
 21. Crowley, P., Huckleberry, N., Biggers, E.: Length-Preserving Encryption with HCTR2. IACR Cryptol. ePrint Arch. p. 1441 (2021), <https://eprint.iacr.org/2021/1441>
 22. Daemen, J., Rijmen, V.: The Design of Rijndael - The Advanced Encryption Standard (AES), Second Edition. Information Security and Cryptography, Springer (2020). <https://doi.org/10.1007/978-3-662-60769-5>, <https://doi.org/10.1007/978-3-662-60769-5>
 23. Datta, N., Dutta, A., Ghosh, S., List, E., Nandi, H.: HCTR+: An Optimally Secure TBC-based Accordion Mode. Cryptology ePrint Archive, Paper 2024/2053 (2024), <https://eprint.iacr.org/2024/2053>
 24. Datta, N., Dutta, A., List, E., Mandal, S.: On the Security of Triplex- and Multiplex-Type Constructions with Smaller Tweaks. In: Chattopadhyay, A., Bhasin, S., Picek, S., Rebeiro, C. (eds.) Progress in Cryptology - INDOCRYPT 2023 - 24th International Conference on Cryptology in India, Goa, India, December 10–13, 2023, Proceedings, Part I. Lecture Notes in Computer Science, vol. 14459, pp. 25–47. Springer (2023). https://doi.org/10.1007/978-3-031-56232-7_2, https://doi.org/10.1007/978-3-031-56232-7_2
 25. Dobraunig, C., Eichlseder, M., Mangard, S., Mendel, F., Mennink, B., Primas, R., Unterluggauer, T.: ISAP v2.0. IACR Trans. Symmetric Cryptol. **2020**(S1), 390–416 (2020). <https://doi.org/10.13154/TOSC.V2020.IS1.390-416>, <https://doi.org/10.13154/tosc.v2020.is1.390-416>
 26. Dobraunig, C., Matusiewicz, K., Mennink, B., Tereschenko, A.: Efficient Instances of Docked Double Decker with AES, and Application to Authenticated Encryption. In: Fehr, S., Fouque, P. (eds.) Advances in Cryptology - EUROCRYPT 2025 - 44th Annual International Conference on the Theory and Applications of Cryptographic Techniques, Madrid, Spain, May 4–8, 2025, Proceedings, Part I. Lecture Notes in Computer Science, vol. 15601, pp. 62–92. Springer (2025). https://doi.org/10.1007/978-3-031-91107-1_3, https://doi.org/10.1007/978-3-031-91107-1_3
 27. Dodis, Y., Pietrzak, K.: Leakage-Resilient Pseudorandom Functions and Side-Channel Attacks on Feistel Networks. In: Rabin, T. (ed.) Advances in Cryptology - CRYPTO 2010, 30th Annual Cryptology Conference, Santa Barbara, CA, USA, August 15–19, 2010. Proceedings. Lecture Notes in Computer Science, vol. 6223, pp. 21–40. Springer (2010). https://doi.org/10.1007/978-3-642-14623-7_2, https://doi.org/10.1007/978-3-642-14623-7_2
 28. Dutta, A., Nandi, M.: Tweakable HCTR: A BBB Secure Tweakable Enciphering Scheme. In: Chakraborty, D., Iwata, T. (eds.) Progress in Cryptology - INDOCRYPT 2018 - 19th International Conference on Cryptology in India, New Delhi, India, December 9–12, 2018, Proceedings. Lecture Notes in Computer Science, vol. 11356, pp. 47–69. Springer (2018). https://doi.org/10.1007/978-3-030-05378-9_3, https://doi.org/10.1007/978-3-030-05378-9_3
 29. Dziembowski, S., Pietrzak, K.: Leakage-Resilient Cryptography. In: 49th Annual IEEE Symposium on Foundations of Computer Science, FOCS 2008, Philadelphia, PA, USA, October 25–28, 2008. pp. 293–302. IEEE Computer

- Society (2008). <https://doi.org/10.1109/FOCS.2008.56>, <https://doi.org/10.1109/FOCS.2008.56>
30. Faust, S., Pietrzak, K., Schipper, J.: Practical Leakage-Resilient Symmetric Cryptography. In: Prouff, E., Schaumont, P. (eds.) *Cryptographic Hardware and Embedded Systems - CHES 2012 - 14th International Workshop*, Leuven, Belgium, September 9-12, 2012. Proceedings. Lecture Notes in Computer Science, vol. 7428, pp. 213–232. Springer (2012). https://doi.org/10.1007/978-3-642-33027-8_13, https://doi.org/10.1007/978-3-642-33027-8_13
 31. Gjiriti, E., Reyhanitabar, R., Vizár, D.: Power Yoga: Variable-Stretch Security of CCM for Energy-Efficient Lightweight IoT. *IACR Trans. Symmetric Cryptol.* **2021**(2), 446–468 (2021). <https://doi.org/10.46586/TOSC.V2021.I2.446-468>, <https://doi.org/10.46586/tosc.v2021.i2.446-468>
 32. Günsing, A., Daemen, J., Mennink, B.: Deck-Based Wide Block Cipher Modes and an Exposition of the Blinded Keyed Hashing Model. *IACR Transactions on Symmetric Cryptology* **2019**(4), 1–22 (Jan 2020). <https://doi.org/10.13154/tosc.v2019.i4.1-22>, <https://tosc.iacr.org/index.php/ToSC/article/view/8451>
 33. Guo, C., Iwata, T., Khairallah, M., Minematsu, K., Peyrin, T.: Romulus v1.3. Submission to NIST Lightweight Cryptography (2021), <https://romulusae.github.io/romulus/docs/Romulusv1.3.pdf>
 34. Guo, C., Iwata, T., Khairallah, M., Minematsu, K., Peyrin, T.: Security Proof for Romulus-T (2022), https://romulusae.github.io/romulus/docs/Romulus_T_proof.pdf
 35. Guo, C., Iwata, T., Minematsu, K.: New Indifferentiability Security Proof of MDPH Hash Function. *IET Inf. Secur.* **16**(4), 262–281 (2022). <https://doi.org/10.1049/ISE2.12058>, <https://doi.org/10.1049/ise2.12058>
 36. Guo, C., Khairallah, M., Peyrin, T.: AET-LR: Rate-1 Leakage-Resilient AEAD based on the Romulus Family (2020), https://thomaspeyrin.github.io/web/assets/docs/slides/Guo-etal-NIST2020_slides.pdf
 37. Guo, C., Pereira, O., Peters, T., Standaert, F.: Authenticated Encryption with Nonce Misuse and Physical Leakage: Definitions, Separation Results and First Construction - (Extended Abstract). In: Schwabe, P., Thériault, N. (eds.) *Progress in Cryptology - LATINCRYPT 2019 - 6th International Conference on Cryptology and Information Security in Latin America*, Santiago de Chile, Chile, October 2-4, 2019, Proceedings. Lecture Notes in Computer Science, vol. 11774, pp. 150–172. Springer (2019). https://doi.org/10.1007/978-3-030-30530-7_8, https://doi.org/10.1007/978-3-030-30530-7_8
 38. Guo, C., Pereira, O., Peters, T., Standaert, F.: Towards Low-Energy Leakage-Resistant Authenticated Encryption from the Duplex Sponge Construction. *IACR Trans. Symmetric Cryptol.* **2020**(1), 6–42 (2020). <https://doi.org/10.13154/TOSC.V2020.I1.6-42>, <https://doi.org/10.13154/tosc.v2020.i1.6-42>
 39. Guo, J., Peyrin, T., Poschmann, A.: The PHOTON Family of Lightweight Hash Functions. In: Rogaway, P. (ed.) *Advances in Cryptology - CRYPTO 2011 - 31st Annual Cryptology Conference*, Santa Barbara, CA, USA, August 14-18, 2011. Proceedings. Lecture Notes in Computer Science, vol. 6841, pp. 222–239. Springer (2011). https://doi.org/10.1007/978-3-642-22792-9_13, https://doi.org/10.1007/978-3-642-22792-9_13
 40. Halevi, S., Rogaway, P.: A Tweakable Enciphering Mode. In: Boneh, D. (ed.) *Advances in Cryptology - CRYPTO 2003, 23rd Annual International Cryptology Conference*, Santa Barbara, California, USA, August 17-21, 2003, Proceedings. Lecture Notes in Computer Science, vol. 2729, pp. 482–499. Springer (2003).

- https://doi.org/10.1007/978-3-540-45146-4_28, https://doi.org/10.1007/978-3-540-45146-4_28
41. Halevi, S., Rogaway, P.: A Parallelizable Enciphering Mode. In: Okamoto, T. (ed.) Topics in Cryptology - CT-RSA 2004, The Cryptographers' Track at the RSA Conference 2004, San Francisco, CA, USA, February 23-27, 2004, Proceedings. Lecture Notes in Computer Science, vol. 2964, pp. 292–304. Springer (2004). https://doi.org/10.1007/978-3-540-24660-2_23, https://doi.org/10.1007/978-3-540-24660-2_23
 42. Hazay, C., López-Alt, A., Wee, H., Wichs, D.: Leakage-Resilient Cryptography from Minimal Assumptions. In: Johansson, T., Nguyen, P.Q. (eds.) Advances in Cryptology - EUROCRYPT 2013, 32nd Annual International Conference on the Theory and Applications of Cryptographic Techniques, Athens, Greece, May 26–30, 2013. Proceedings. Lecture Notes in Computer Science, vol. 7881, pp. 160–176. Springer (2013). https://doi.org/10.1007/978-3-642-38348-9_10, https://doi.org/10.1007/978-3-642-38348-9_10
 43. Jean, J., Nikolic, I., Peyrin, T., Seurin, Y.: The Deoxys AEAD Family. J. Cryptol. **34**(3), 31 (2021). <https://doi.org/10.1007/S00145-021-09397-W>, <https://doi.org/10.1007/s00145-021-09397-w>
 44. Jimale, M.A., Abdullah, N.A., Kiah, M.L.M., Idris, M.Y.I.B., Z'aba, M.R., Jamil, N., Rohmad, M.S.: Sponge-Based Parallel Authenticated Encryption With Variable Tag Length and Side-Channel Protection. IEEE Access **11**, 59661–59674 (2023). <https://doi.org/10.1109/ACCESS.2023.3267161>, <https://doi.org/10.1109/ACCESS.2023.3267161>
 45. Khairallah, M.: Security of COFB against Chosen Ciphertext Attacks. IACR Trans. Symmetric Cryptol. **2022**(1), 138–157 (2022). <https://doi.org/10.46586/TOSC.V2022.I1.138-157>, <https://doi.org/10.46586/tosc.v2022.i1.138-157>
 46. Khairallah, M.: A note on -Tweakable HCTR: A BBB Secure Tweakable Enciphering Scheme-. IACR Cryptol. ePrint Arch. p. 600 (2024), <https://eprint.iacr.org/2024/600>
 47. Khairallah, M.: CCA Security with Short AEAD Tags. IACR Commun. Cryptol. **1**(1), 11 (2024). <https://doi.org/10.62056/AEVUA69P1>, <https://doi.org/10.62056/aevua69p1>
 48. Khairallah, M.: Revisiting Leakage-Resilient MACs and Succinctly-Committing AEAD More Applications of Pseudo-Random Injections. IACR Trans. Symmetric Cryptol. **2025**(1), 211–239 (2025). <https://doi.org/10.46586/TOSC.V2025.I1.211-239>, <https://doi.org/10.46586/tosc.v2025.i1.211-239>
 49. Kumar, M.: Security of XCB and HCTR. CoRR **abs/2308.06082** (2023). <https://doi.org/10.48550/ARXIV.2308.06082>, <https://doi.org/10.48550/arXiv.2308.06082>
 50. Lapid, B., Wool, A.: Cache-Attacks on the ARM TrustZone Implementations of AES-256 and AES-256-GCM via GPU-Based Analysis. In: Cid, C., Jr., M.J.J. (eds.) Selected Areas in Cryptography - SAC 2018 - 25th International Conference, Calgary, AB, Canada, August 15-17, 2018, Revised Selected Papers. Lecture Notes in Computer Science, vol. 11349, pp. 235–256. Springer (2018). https://doi.org/10.1007/978-3-030-10970-7_11, https://doi.org/10.1007/978-3-030-10970-7_11
 51. Liskov, M.D., Rivest, R.L., Wagner, D.A.: Tweakable Block Ciphers. In: Yung, M. (ed.) Advances in Cryptology - CRYPTO 2002, 22nd Annual International Cryptology Conference, Santa Barbara, California, USA, August 18-22, 2002, Proceedings. Lecture Notes in Computer Science, vol. 2442, pp. 31–

46. Springer (2002). https://doi.org/10.1007/3-540-45708-9_3, https://doi.org/10.1007/3-540-45708-9_3
52. List, E.: TEDT2 - Highly Secure Leakage-Resilient TBC-Based Authenticated Encryption. In: Longa, P., Ràfols, C. (eds.) Progress in Cryptology - LATIN-CRYPT 2021 - 7th International Conference on Cryptology and Information Security in Latin America, Bogotá, Colombia, October 6-8, 2021, Proceedings. Lecture Notes in Computer Science, vol. 12912, pp. 275–295. Springer (2021). https://doi.org/10.1007/978-3-030-88238-9_14, https://doi.org/10.1007/978-3-030-88238-9_14
53. Luby, M., Rackoff, C.: How to Construct Pseudorandom Permutations from Pseudorandom Functions. SIAM Journal on Computing **17**(2), 373–386 (1988), <https://epubs.siam.org/doi/abs/10.1137/0217022>
54. Martin, D.P.: Leakage-Resilient Symmetric Cryptography: Theory and Practice. Ph.D. thesis, University of Bristol, UK (2016), <https://ethos.bl.uk/OrderDetails.do?uin=uk.bl.ethos.702904>
55. McGrew, D.A., Viega, J.: The Security and Performance of the Galois/Counter Mode (GCM) of Operation. In: Canteaut, A., Viswanathan, K. (eds.) Progress in Cryptology - INDOCRYPT 2004, 5th International Conference on Cryptology in India, Chennai, India, December 20-22, 2004, Proceedings. Lecture Notes in Computer Science, vol. 3348, pp. 343–355. Springer (2004). https://doi.org/10.1007/978-3-540-30556-9_27, https://doi.org/10.1007/978-3-540-30556-9_27
56. McKay, K., Bassham, L., Sönmez Turan, M., Mouha, N.: Report on Lightweight Cryptography. Tech. rep., National Institute of Standards and Technology (2016), https://csrc.nist.gov/CSRC/media/Publications/nistir/8114/draft/documents/nistir_8114_draft.pdf
57. Mennink, B.: XPX: Generalized Tweakable Even-Mansour with Improved Security Guarantees. In: Robshaw, M., Katz, J. (eds.) Advances in Cryptology - CRYPTO 2016 - 36th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 14-18, 2016, Proceedings, Part I. Lecture Notes in Computer Science, vol. 9814, pp. 64–94. Springer (2016). https://doi.org/10.1007/978-3-662-53018-4_3, https://doi.org/10.1007/978-3-662-53018-4_3
58. Mennink, B.: Understanding the Duplex and Its Security. IACR Trans. Symmetric Cryptol. **2023**(2), 1–46 (2023). <https://doi.org/10.46586/TOSC.V2023.I2.1-46>, <https://doi.org/10.46586/tosc.v2023.i2.1-46>
59. Mohajerani, K., Beckwith, L., Abdulgadir, A., Ferrufino, E., Kaps, J.P., Gaj, K.: SCA Evaluation and Benchmarking of Finalists in the NIST Lightweight Cryptography Standardization Process. Cryptology ePrint Archive, Paper 2023/484 (2023), <https://eprint.iacr.org/2023/484>
60. Momin, C., Standaert, F., Verhamme, C.: Primitive-Level vs. Implementation-Level DPA Security: a Certified Case Study (Pleading for Standardized Leakage-Resilient Cryptography). IACR Trans. Cryptogr. Hardw. Embed. Syst. **2025**(3), 717–744 (2025). <https://doi.org/10.46586/TCHES.V2025.I3.717-744>, <https://doi.org/10.46586/tches.v2025.i3.717-744>
61. Safavi-Naini, R., Lisý, V., Desmedt, Y.: Economically Optimal Variable Tag Length Message Authentication. In: Kiayias, A. (ed.) Financial Cryptography and Data Security - 21st International Conference, FC 2017, Sliema, Malta, April 3-7, 2017, Revised Selected Papers. Lecture Notes in Computer Science, vol. 10322, pp. 204–223. Springer (2017). https://doi.org/10.1007/978-3-319-70972-7_11, https://doi.org/10.1007/978-3-319-70972-7_11

62. Shen, Y., Peters, T., Standaert, F.: Multiplex: TBC-Based Authenticated Encryption with Sponge-Like Rate. *IACR Trans. Symmetric Cryptol.* **2024**(2), 1–34 (2024). <https://doi.org/10.46586/TOSC.V2024.I2.1-34>, <https://doi.org/10.46586/tosc.v2024.i2.1-34>
63. Shen, Y., Peters, T., Standaert, F., Cassiers, G., Verhamme, C.: Triplex: an Efficient and One-Pass Leakage-Resistant Mode of Operation. *IACR Trans. Cryptogr. Hardw. Embed. Syst.* **2022**(4), 135–162 (2022). <https://doi.org/10.46586/TCHES.V2022.I4.135-162>, <https://doi.org/10.46586/tches.v2022.i4.135-162>
64. Shrimpton, T., Terashima, R.S.: A Modular Framework for Building Variable-Input-Length Tweakable Ciphers. In: Sako, K., Sarkar, P. (eds.) *Advances in Cryptology - ASIACRYPT 2013 - 19th International Conference on the Theory and Application of Cryptology and Information Security, Bengaluru, India, December 1-5, 2013, Proceedings, Part I. Lecture Notes in Computer Science*, vol. 8269, pp. 405–423. Springer (2013). https://doi.org/10.1007/978-3-642-42033-7_21, https://doi.org/10.1007/978-3-642-42033-7_21
65. Standaert, F., Pereira, O., Yu, Y.: Leakage-Resilient Symmetric Cryptography under Empirically Verifiable Assumptions. In: Canetti, R., Garay, J.A. (eds.) *Advances in Cryptology - CRYPTO 2013 - 33rd Annual Cryptology Conference, Santa Barbara, CA, USA, August 18-22, 2013. Proceedings, Part I. Lecture Notes in Computer Science*, vol. 8042, pp. 335–352. Springer (2013). https://doi.org/10.1007/978-3-642-40041-4_19, https://doi.org/10.1007/978-3-642-40041-4_19
66. Standaert, F., Pereira, O., Yu, Y., Quisquater, J., Yung, M., Oswald, E.: Leakage Resilient Cryptography in Practice. In: Sadeghi, A., Naccache, D. (eds.) *Towards Hardware-Intrinsic Security - Foundations and Practice*, pp. 99–134. *Information Security and Cryptography*, Springer (2010). https://doi.org/10.1007/978-3-642-14452-3_5, https://doi.org/10.1007/978-3-642-14452-3_5
67. Wang, P., Feng, D., Wu, W.: HCTR: A Variable-Input-Length Enciphering Mode. In: Feng, D., Lin, D., Yung, M. (eds.) *Information Security and Cryptology, First SKLOIS Conference, CISC 2005, Beijing, China, December 15-17, 2005, Proceedings. Lecture Notes in Computer Science*, vol. 3822, pp. 175–188. Springer (2005). https://doi.org/10.1007/11599548_15, https://doi.org/10.1007/11599548_15

A Leakage Functions

Throughout the paper, we refer to the leakage function of different primitives, and different leakage models. These models can be summarized by three concepts:

- Leak-free functions: a function or a primitive is leak-free if an adversary/observer cannot learn anything about its execution except what they can infer from its inputs and outputs defined in its syntax and revealed by the oracle. In other words, if the oracle executes a leak-free function and the inputs and/or outputs are only used internally by the oracle, then the adversary learns nothing about the function execution, except what they can infer from the interaction with the oracle.
- Leakage-resilient functions: a function or a primitive is leakage-resilient if its execution leaks some information that is not inferred from the oracle interaction, but is assumed to still be secure despite the leaked information. The leaked information can be defined by an associated leakage function if needed. Leakage resilience of a primitive is typically an assumption: we assume that a TBC is still unpredictable even with leakage, or we assume that a value comparison function is still secure with leakage. In this case, the security is captured by a notion that embodies this assumption. For larger constructions, we need a security proof that reduces the security to smaller primitive that are themselves either leak-free, leakage-resilient or have unbounded leakage. For instance, we can assume that a block cipher has a non-trivial leakage functions. This makes it harder to assume it is indistinguishable from a random permutation, but it maintains a security property such as unpredictability and/or resistance against key recovery despite the leakage.
- Functions with unbounded leakage: These functions are assumed to leak all their internal values, execution flow, etc. Formally, some functions appear in the security proof as monolithic functions where the internals are not specified, but saying a function has unbounded leakage means that we make no assumptions at all about what the adversary can learn when the function is executed. Note that this also means that even if the function's inputs and outputs are hidden inside the oracle, the adversary is assumed to learn them if the function has unbounded leakage.

We refer to [8] for a more in-depth discussion on the different levels and types of leakage assumptions in this context.

B Deferred Proofs

B.1 Proof of Theorem 1

Assume, without loss of generality, that $\mathcal{B}$ terminates after the first successful forgery. $\mathcal{A}$ runs $\mathcal{B}$. If $\mathcal{B}$ makes an encryption query, $\mathcal{A}$ makes the non-challenge

encryption query and returns the response and corresponding leakage to $\mathcal{B}$. If $\mathcal{B}$ makes a forgery query, $\mathcal{A}$ first makes a (leak-free) challenge decryption query. Then, it decodes the response. If the response is a valid codeword, $\mathcal{A}$ returns 1 and the game ends. If it is not a valid codeword, then $\mathcal{A}$ sends $\perp$ to $\mathcal{B}$. In the real world, the probability that $\mathcal{A}$ returns 1 is $\text{Adv}_{\text{EtE}[\delta, \tilde{H}]}^{\text{CIML1}}(\mathcal{B})$. In the ideal world, the probability that $\mathcal{A}$ returns 1 is at most 2δ .

$$\text{Adv}_{\tilde{H}}^{\text{sTPRP12}}(\mathcal{A}) \geq \text{Adv}_{\text{EtE}[\delta, \tilde{H}]}^{\text{CIML1}}(\mathcal{B}) - \text{Adv}_{\text{EtE}[\delta, \tilde{R}]}^{\text{CIML1}}(\mathcal{B})$$

or

$$\text{Adv}_{\text{EtE}[\delta, \tilde{H}]}^{\text{CIML1}}(\mathcal{B}) \leq \text{Adv}_{\tilde{H}}^{\text{sTPRP12}}(\mathcal{A}) + 2q_f\delta.$$

The last term follows from the security of the EtE framework in the ideal world, and a hybrid argument over forgery attempts.

B.2 Proof of Theorem 4

Let $\mathcal{A}$ be an adversary. At the beginning of the game, $\mathcal{A}$ chooses a random bit b and runs $\mathcal{B}$. When $\mathcal{B}$ makes an encryption non-challenge query, $\mathcal{A}$ codes the message and forwards it to its leaky encryption oracle, then forwards the response to $\mathcal{B}$. When $\mathcal{B}$ makes a decryption non-challenge query, $\mathcal{A}$ makes the corresponding leaky decryption query, and checks whether the response is a valid codeword. If it is, $\mathcal{A}$ forwards the decoded response to $\mathcal{B}$. Otherwise, it sends $\perp$. In all the previous cases, $\mathcal{A}$ also sends the corresponding leakage to $\mathcal{B}$. When $\mathcal{B}$ makes a challenge query (N, A, M_0, M_1) , $\mathcal{A}$ makes the query $(N \| A, M_b)$ to its encryption challenge oracle, and forwards the response back to $\mathcal{B}$. At the end of the game, $\mathcal{B}$ returns b' . $\mathcal{A}$ checks if $b = b'$ and returns 1 if so.

In the real world, since $\mathcal{B}$ can guess b with a certain probability, $\mathcal{A}$ outputs 1 with the same probability.

$$\begin{aligned} \Pr[\mathcal{A} \Rightarrow 1 \text{ in the real world}] &= \Pr[b = b' \text{ in the real world}] = \\ &\quad \text{Adv}_{\text{EtE}[\delta, \tilde{H}]}^{\text{CCAM}\ell 2}(\mathcal{B}) + 1/2. \end{aligned}$$

Instead of calculating the probability that $\mathcal{B}$ guesses b correctly in the ideal world, we do so in a temporary world, then show that the two worlds are indistinguishable. Instead of the random tweakable permutation used for the challenge oracles, the challenge oracles simply output random string of appropriate length. In this world, $\mathcal{B}$ receives random strings that are independent of $\tilde{H}$ and b . Thus,

$$\Pr[\mathcal{A} \Rightarrow 1 \text{ in the temporary world}] = \Pr[b = b' \text{ in the temporary world}] = 1/2.$$

We the advantage of $\mathcal{A}$ against distinguishing between this temporary world and the real world as

$$\begin{aligned} &\text{Adv}_{\tilde{H}}^{\text{sTPRP}\ell 2^*}(\mathcal{A}) \\ &= |\Pr[\mathcal{A} \Rightarrow 1 \text{ in the real world}] - \Pr[\mathcal{A} \Rightarrow 1 \text{ in the temporary world}]| \\ &\geq \text{Adv}_{\text{EtE}[\delta, \tilde{H}]}^{\text{CCAM}\ell 2}(\mathcal{B}), \end{aligned}$$

or

$$\text{Adv}_{\text{EtE}[\delta, \tilde{H}]}^{\text{CCAM}\ell 2}(\mathcal{B}) \leq \text{Adv}_{\tilde{H}}^{\text{STPRP}^*}(\mathcal{A}).$$

Next, we apply a standard hybrid argument to replace the third and fourth oracles of $\mathcal{A}$ with a tweakable permutation selected uniformly at random. This change is a standard PRP-PRF switch with transition probability

$$\frac{q(q-1)}{2^{\min(|M|)+1}}.$$

B.3 Proof of Theorem 6

We build a sequence of hybrid games. Let E_i be the event that the adversary forges against game G_i . W.l.o.g., we assume $\mathcal{A}$ terminates after the first successful forgery, if any.

G0: The real-world construction with early abort: during a forgery attempt, the verifier checks if the first t bits of the output of X are equal to the first t bits of C_r . If not, it terminates the algorithm.

G1: We replace the leak-free TBC $\tilde{E}_K$ with a TURP. There exists an adversary $\mathcal{A}_1$, such that

$$|\Pr[E_0] - \Pr[E_1]| \leq \text{Adv}_{\tilde{E}}^{\text{STPRP}}(\mathcal{A}_1).$$

G2: We consider an adversary $\mathcal{A}_2$ that has access to the construction, and forwards $\mathcal{A}$'s queries to the construction and sends back the responses to $\mathcal{A}$. For every encryption query, $\mathcal{A}_2$ makes two queries to the hash function H : $H(N||A||A||0^t||M_r)$ and $H(N||A||A||C_r)$ and adds the input-output pair to a hash table $\mathcal{H}$. For every forgery attempt, $\mathcal{A}_2$ makes a query $H(N||A||A||C_r)$ and adds it to $\mathcal{H}$. At the end of the game, $\mathcal{A}_2$ checks $\mathcal{H}$ and terminates the game if it includes a collision. From the assumption on H ,

$$|\Pr[E_1] - \Pr[E_2]| \leq \epsilon_{\text{CR}}.$$

where H is $(t', \epsilon_{\text{CR}})$ -collision-resistant, such that $t' > t + (2q_e + q_f)t_H$, where t_H is the time of one call to H . In the remainder of the proof, we set T_w to be the input to H for each hash query.

G3: We replace the composition of H and the TURP with a function $\tilde{F}$ that operates as follows: For each forward query (T_w, Y) (resp. backward query (T_w, Z)), it checks if there is a previous query such that $\tilde{F}(T_w, Y) = Z$, in which case it returns Z (resp. Y). Otherwise, it returns a uniformly random values. This transition is covered by the result of [40, Lemma 1], with

$$|\Pr[E_2] - \Pr[E_3]| \leq \frac{(2q_e + q_f)^2}{2^n}.$$

G4: We terminate the game if there exists two queries to $\tilde{F}$ such that $\tilde{F}(T_w, Y) = \tilde{F}(T'_w, Y')$ such that $(T_w, Y) \neq (T'_w, Y')$, or $\tilde{F}^{-1}(T_w, Z) = \tilde{F}^{-1}(T'_w, Z')$ such that $(T_w, Z) \neq (T'_w, Z')$. The probability of this event is at most $\binom{2q_e + q_f}{2}/2^n$. Thus,

$$|\Pr[E_3] - \Pr[E_4]| \leq \frac{\binom{2q_e + q_f}{2}}{2^n}.$$

We note that this transition is needed although some of the collisions are already covered in the previous transition. This is because the previous transition only deals with collisions of the form $\tilde{F}(T_w, Y) = \tilde{F}(T_w, Y')$ or $\tilde{F}^{-1}(T_w, Z) = \tilde{F}^{-1}(T_w, Z')$, but does not deal with collisions that involve different tweaks.

G5: This game is constructed such that the inputs to X are: (a) uniformly distributed, and (b) collision free. Let $i \leq q_f$ be the i^{th} forgery attempt. We define the bad event:

- B5a: there exists an encryption query (N, A, M_l, M_r) where

$$(N_i, A_i, \theta_i, C_{r_i}) = (N, A, M_l, M_r).$$

- B5b: Let θ_i be the input to X in the i^{th} forgery attempt. $\exists 1 < j < q_p$, s.t. $\theta_j = \theta_i$, where the adversary has made the offline query (θ_j, N) to X_P prior to the i^{th} forgery attempt.

The game terminates in any of these events occurs. If either B5a occurs, then there is an adversary $\mathcal{A}_3$ against the GuessL1 security of X . B5a means that B5a occurs at for the i^{th} forgery attempt. $\mathcal{A}_3$ has offline access to X , and can simulate $\tilde{F}$. $\mathcal{A}_3$ simulates G5, except it replaces one of the simulated queries to X with the query to its X GuessL1 challenger. If B5a occurs in this simulated game, then the correct guess for the replaced query is M_l .

The analysis of B5a game is similar to the analysis of G5 in the proof of Theorem 5, where the GuessL1 replaces one of the encryption queries to include the challenge, and outputs guesses based on the forgery attempts. On the other hand, B5b can only occur by a random collision, as θ_i is sampled randomly since the adversary has no leakage corresponding to the forgery attempt before generating it. Thus,

$$|\Pr[E_4] - \Pr[E_5]| \leq \frac{q_f q_p}{2^n} + 2q_e \text{Adv}_X^{\text{GuessL1}[q_f]}(\mathcal{A}_3) + \frac{(q_e + q_f)^2}{2^n}.$$

G6: We replace the first t -bit outputs of X during the forgery attempts with random strings. We can construct a PRGL1 adversary $\mathcal{A}_4$ that returns 1 if and only if any forgery attempt is successful. We replace all instances of X during encryption and forgery queries such that X is replaced by a random string. Note that we still return the leakage from X . This is equivalent to $\mathbb{S}_L$, thus,

$$|\Pr[E_5] - \Pr[E_6]| \leq \text{Adv}_X^{\text{PRGL1}}(\mathcal{A}_4),$$

where $\mathcal{A}_4$ makes $q_e + q_f$ challenge queries and q_p offline queries to X_P . Finally, the probability that the i^{th} forgery attempt is successful is 2^{-t} , since the first t bits are uniformly distributed. Thus,

$$\Pr[E_6] = \frac{q_f}{2^t}.$$

B.4 Proof of Theorem 7

We construct a sequence of hybrid games. We define E_i as the event that $\mathcal{A}$ returns 1 in game G_i . We require that all non-trivial challenge queries return $\perp$ and terminate early. Thus, the proof of Theorem 6 is repeated, giving the transition:

$$|\Pr[E_0] - \Pr[E_1]| = \text{Adv}^{\text{CIML2}}(\mathcal{A}_1).$$

where G_1 is G_6 in the proof of Theorem 6.

G2: We terminate the game if during any encryption query θ_i is the input to X , and an offline query has been made to X with the same value. At the i^{th} query, the probability of this event is bounded by the probability that an adversary guesses the input by outputting the offline queries that $\mathcal{A}_3$ makes. By applying a hybrid argument similar to the proof of Theorem 6 over all encryption queries.

$$|\Pr[E_1] - \Pr[E_2]| \leq (q_e + q_c) \text{Adv}^{\text{GuessL1}[q_p]}(\mathcal{A}_2).$$

G3: We replace the outputs of X during the challenge queries with random strings. We can construct a **PRGL1** adversary $\mathcal{A}_3$ that returns 1 if and only if $\mathcal{A}$ returns 1. $\mathcal{A}_3$ makes $q_e + q_c + q_f$ challenge queries and q_p offline queries to X_P , such that

$$|\Pr[E_2] - \Pr[E_3]| \leq \text{Adv}_X^{\text{PRGL1}}(\mathcal{A}_3).$$

G4: In this game, we use the combined **LORL2** and the fact that during challenge queries the nonce is always fresh. We go through the challenge queries one by one, swapping $M_0^{(i)}$ and $M_1^{(i)}$. Since the nonce is fresh, and the hash function is collision free, θ_i is sampled uniformly at random, independent of $M_{l_0}^{(i)}$ and $M_{l_1}^{(i)}$. Thus, the adversary can only win by distinguishing between $M_{r_0}^{(i)}$ and $M_{r_1}^{(i)}$. The distinguishing probability is bounded by

$$\text{Adv}_{\text{comb}}^{\text{LORL2}}(\mathcal{A}_4^i)$$

where $\mathcal{A}_4^i$ is the adversary that simulates G_4 and replaces the right part of the i^{th} challenge query with its challenge. By applying a similar hybrid argument as in the previous game,

$$|\Pr[E_3] - \Pr[E_4]| \leq \sum_{i=1}^{q_c} \text{Adv}_{\text{comb}}^{\text{LORL2}}(\mathcal{A}_4^i) \leq q_c \text{Adv}_{\text{comb}}^{\text{LORL2}}(\mathcal{A}_4)$$

where $\mathcal{A}_4 \equiv \mathcal{A}_4^i$ who has the maximum advantage. By swapping all the messages, eventually $b = 0$ and $b = 1$ are indistinguishable. This concludes the proof.

B.5 Security proof of Alicante

In G_0 , the challenger implements Algorithm 3. In G_1 , we replace all the calls where the first parameter of the tweak is 0 and 3 by π_0 and π_3 , respectively. The transition advantage is bounded by $q_p/2^k$, according to Definition 2.

G_2 : Similar to the proof of Theorem 5, we now simulate the challenger with oracle access to π_0, π_3 , and the ideal cipher. We terminate the game if a collision is found at the output of MDPH. The transition advantage follows from [35, Theorem 3.1]. Let the total number of construction queries be q , the total number of $(t_w - n)$ -bit blocks processed by H be σ_H , then the probability of such collision occurring is upper bounded by

$$\frac{6(6\sigma_H + (3 + n)q_p)}{(2^n - 4\sigma_H - 2q_p)^2} + \frac{24n\sigma_H + 12nq_p}{2^n - 4\sigma_H - 2q_p} + \frac{(nq_p + 2q)^2}{2^{2n}}.$$

Note that each construction calls includes two hash calls, and double the number of blocks.

G_3 and G_4 are identical to the corresponding game in the proof of Theorem 5, with transition advantage bounded by

$$\frac{8q^2}{2^n}.$$

G_5, G_6, G_7 : The analysis of these transitions follows from Yu *et al.* [10, Lemma 6]. The transition advantage to G_5 is bounded by

$$\frac{q_e + q_d}{2^n} + (q_e + q_d)\text{Adv}_{\tilde{E}}^{2\text{-up}[q_g]}(\mathcal{A}_1)$$

$q_g = q_{IC} + 4\sigma_h$: q_{IC} is the number of offline ideal cipher calls, and σ_H is the number of blocks absorbed by MDPH. The transition advantage to G_6 is bounded by

$$\frac{(q_{ce} + q_{cd})q_p}{2^n}.$$

The transition advantage to G_7 is bounded by

$$(2\sigma_X)\text{Adv}_{\tilde{E}}^{2\text{-up}[q_g]}(\mathcal{A}_2).$$

where σ_X is the number of n -bit blocks of M_r in all queries. The rest of the proof follows from the proof of Theorem 5.

Comments on the proof. This assumption is practical and is common in the proofs of leveled schemes. For instance, the authors of [10] use exactly the same assumption in their analysis where they assume that the keyed, leak-free instances of the TBC can be replaced by ideal random tweakable permutations. In their case, if the key appears in any of the queries to other TBC calls it can be problematic, which leads to a increasing the offline power of the adversaries. We avoid this by domain separation, where even if the key value appeared in a different query, the domains will be different, and the adversary cannot detect such occurrence using the queries. In principle, the adversary can still detect this occurrence, for example by matching the leakages from the key scheduling function. This is not possible if we restrict The usage of static secret keys to the leak-free domain. Under the ideal cipher assumption, replacing $\tilde{E}_K^{0,T}(M)$ by

$\pi_0(K, T, M)$ can be done with a hybrid argument, where the distinguishing game is terminated if the adversary uses the key in any offline query to the ideal cipher, which occurs with probability at most $p/2^k$ after p queries to the ideal cipher. Such assumption is referred to in [10] as the oracle-free leakage assumption; the leakage from one oracle query should not leak the response to a different oracle query.

In the next part of the analysis, we model the leakage function when $B = 1$, such that it leaks all the ideal cipher inputs and outputs. Alternatively, we define the leakage function when $B = 2$ according to the non-invertible leakage assumption of [10, Section 6]. In fact, we use exactly the same stream cipher as [10, Section 6]. Thus, [10, Lemma 6] applies to the middle layer of our design. This gives us a $\text{PRG}\ell 1$ term in the order of $O(D^2/(c \cdot 2^n))$, where D is the total data complexity counted in blocks and c is a small constant based on the side-channel protection of $\tilde{E}$ when $B = 1$. The non-invertible leakage assumption on the ideal cipher in the 2PRG scheme requires that any key to the ideal cipher appears in only two leaky queries. As we argued in the description **Qaitbay**, the generic analysis of **MaB** does not hold directly as the ideal cipher may leak useful information about the keys even if the tweaks are different. Thus, during the **GuessL1**, $\text{PRG}\ell 1$ and PRGL1 games, we have to consider the total number of queries made to the ideal cipher and not just the ideal cipher queries in X .

Finally, we rely on the independence between the ideal cipher calls with $B = 1$ vs $B = 2$, and instantiate H using the **MDPH** hash function [35]. Now that the rest of the analyses follows exactly from the proof of the **Motte-and-Bailey** framework.

C Diagrams of Alicante

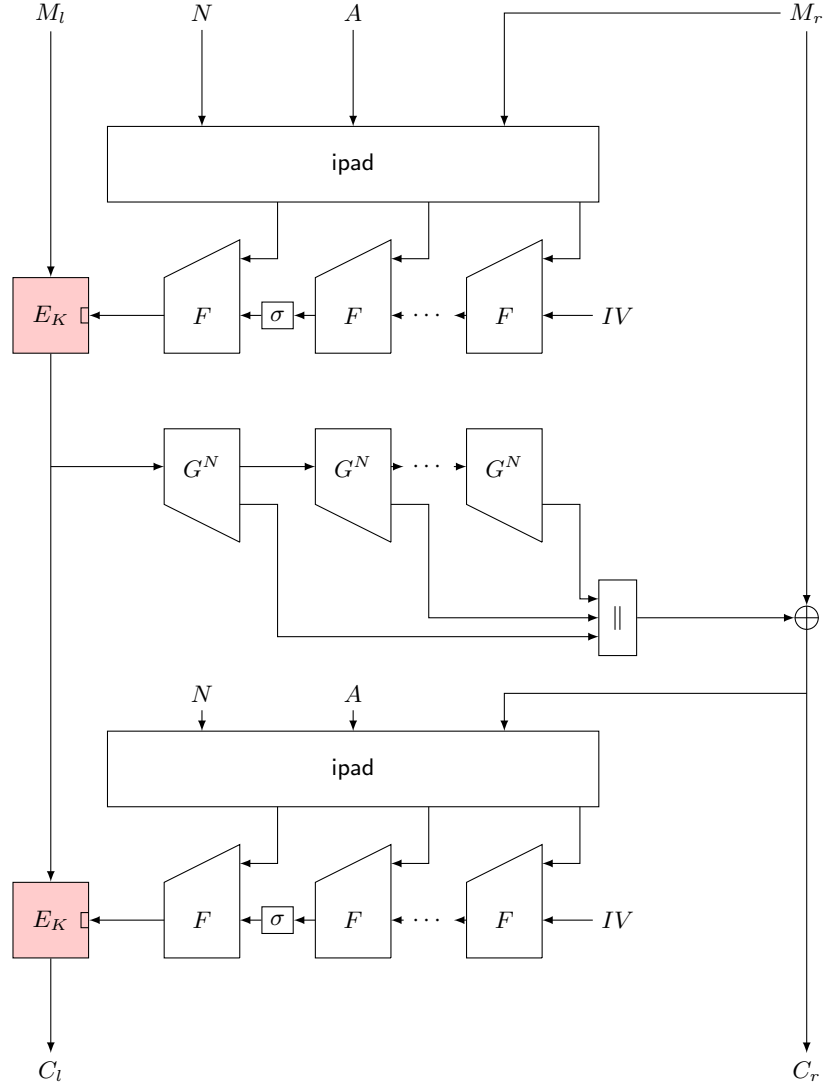


Fig. 2: The **Alicante-1**: F is Hirose's compression function and G is one step of 2PRG.

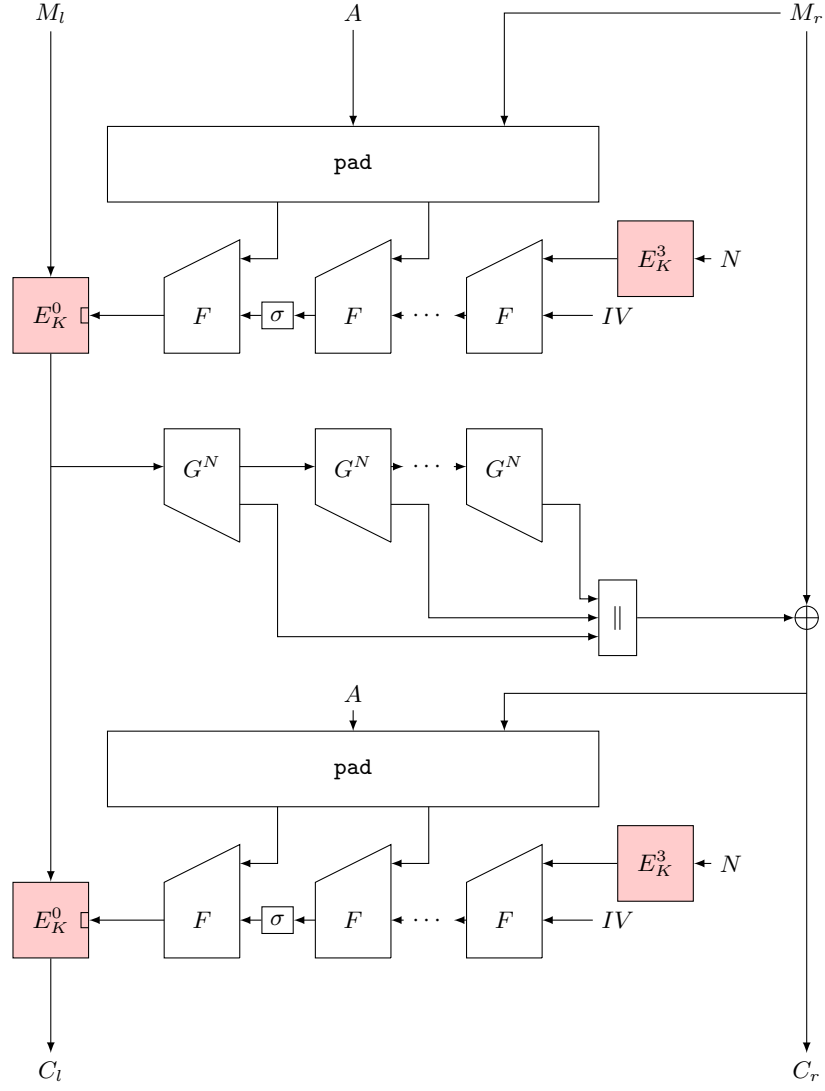


Fig. 3: The **Alicante-2**: F is Hirose's compression function and G is one step of 2PRG.